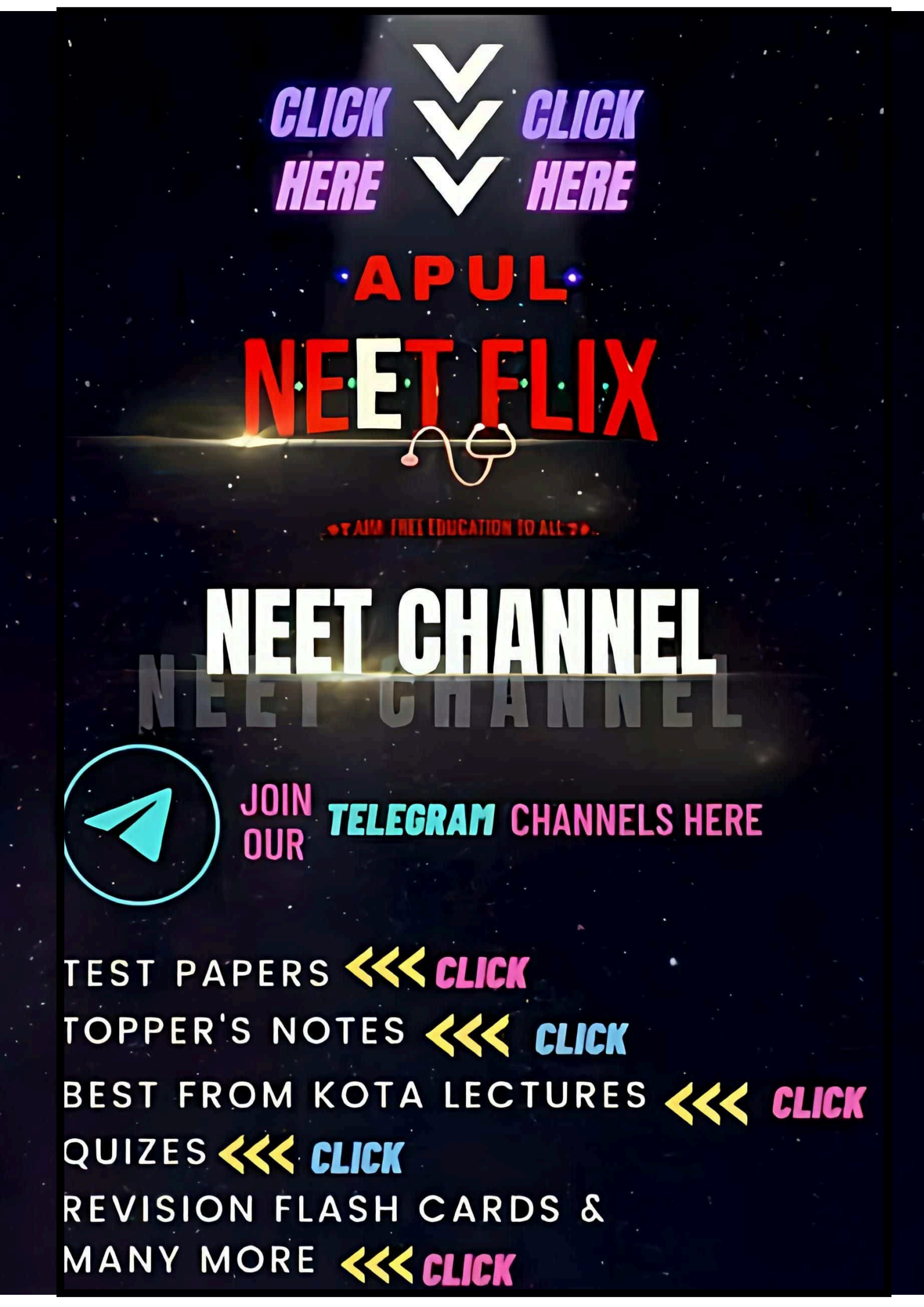

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Body Fluids and Circulation

Q.NO.	QUESTIONS	ANSWERS
1.	All living cells have to be provided with _____, _____ and other essential substances.	nutrients, O_2
2.	The _____ or _____ produced, have to be _____ for healthy functioning of tissues.	Waste, harmful substances, removed continuously
3.	Different groups of animals have evolved _____ for this transport.	Different
4.	Simple organisms like _____ and _____ circulate water from their surroundings through their _____ to facilitate the cells to exchange these substances.	Sponges, coelenterates, body cavities
5.	More complex organisms use _____ within their bodies to transport such materials.	special fluids
6.	_____ is the most commonly used body fluid by most of the higher organisms including humans for this purpose.	Blood
7.	Another body fluid, _____ also helps in the transport of certain substances.	Lymph
	BLOOD	
8.	Blood is a _____ consisting of a _____ plasma, and formed elements.	special connective tissue, fluid matrix
9.	Plasma is a _____ coloured, _____ fluid constituting nearly _____ of the blood.	Straw, viscous, 55%
10.	_____ per cent of plasma is water .	90-92
11.	proteins contribute _____ per cent of plasma.	6-8
12.	_____, _____ and albumins are the major proteins.	Fibrinogen, globulins
13.	Fibrinogens are needed for _____ or _____ of blood.	Clotting, coagulation
11.	Globulins primarly are involved in _____.	defense mechanisms of the body
12.	the albumins help in _____.	osmotic balance
13.	Plasma also contains small amounts of minerals like Na^+, _____, _____, _____, Cl^-, etc.	Ca^{++}, Mg^{++}, HCO_3^-
14.	Glucose, _____, _____, etc., are also present in the plasma as they are always _____ in the body.	amino acids, lipids, in transit
15.	Factors for coagulation or clotting of blood are also present in the plasma in _____.	An inactive form
16.	Plasma without the _____ is called serum.	clotting factors

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	FORMED ELEMENTS	
17.	Erythrocytes, _____ and _____ are collectively called formed elements and they constitute nearly _____ of the blood.	Leucocytes, platelets, 45 percent
18.	_____ are the most abundant of all the cells in blood. A healthy adult man has, on an average, _____ of RBCs mm^{-3} of blood.	Erythrocytes or red blood cells (RBC) 5 million to 5.5 million
19.	RBCs are formed in the _____ in the adults.	red bone marrow
20.	RBCs are devoid of _____ in most of the mammals and are _____ in shape.	Nucleus, biconcave
21.	They have a _____, iron containing _____ called haemoglobin, hence the colour and name of these cells.	red coloured, complex protein
22.	A healthy individual has _____ gms of haemoglobin in every 100 ml of blood. These molecules play a significant role in _____.	12-16, transport of respiratory gases
23.	RBCs have an average life span of _____ after which they are destroyed in the _____ (graveyard of RBCs).	120 days, spleen
24.	_____ are also known as white blood cells (WBC) as they are_____ due to the lack of haemoglobin.	Leucocytes, colourless
25.	They are _____ and are relatively _____ in number which averages _____ mm^{-3} of blood.	Nucleated, lesser, 6000-8000
26.	Leucocytes are generally _____ lived.	Short
27.	We have two main categories of WBCs –_____ and _____.	granulocytes, agranulocytes
28.	_____, _____ and _____ are different types of granulocytes, while _____ and _____ are the agranulocytes.	Neutrophils, eosinophils, basophils, lymphocytes, monocytes
29.	_____ are the most abundant cells (60-65 per cent) of the total WBCs and _____ are the least (0.5-1 per cent) among them. _____ and _____ (6-8 per cent) are phagocytic cells which destroy foreign organisms entering the body.	Neutrophils, basophils, Neutrophils, monocytes
30.	_____ secrete histamine, serotonin, heparin, etc., and are involved in inflammatory reactions.	Basophils
31.	_____ (2-3 per cent) resist infections and are also associated with _____.	Eosinophils
32.	Lymphocytes (20-25 per cent) are of two major types – _____ and _____ forms.	'B', 'T'
33.	_____ and _____ are responsible for immune responses of the body.	B, T lymphocytes
34.	Platelets also called _____ are cell fragments produced from _____ (special cells in the bone marrow).	thrombocytes, megakaryocytes
35.	Blood normally contains _____ platelets mm^{-3}.	1,500,00-3,500,00

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36.	Platelets can release a variety of substances most of which are involved in the _____ or _____. A reduction in their number can lead to _____ which will lead to _____ from the body.	Coagulation, clotting, clotting disorders, excessive loss of blood
	BLOOD GROUPS	
37.	Two such groupings – the _____ and _____ are widely used all over the world.	ABO, Rh
38.	ABO grouping is based on the presence or absence of _____ (chemicals that can induce immune response) on the RBCs.	two surface antigens
39.	The plasma of different individuals contain _____ natural antibodies (proteins produced in response to antigens).	Two
40.	During blood transfusion, _____ blood cannot be used; the blood of a donor has to be _____ with the blood of a recipient before any blood transfusion to avoid _____.	Any, carefully matched, severe problems of clumping (destruction of RBC)
41.	It is evident that group _____ blood can be donated to persons with any other blood group and _____ group individuals are called 'universal donors'.	O, O
42.	Persons with _____ group can accept blood from persons with ____ as well as the other groups of blood.	AB, AB
43.	An Rh _____ person, if exposed to Rh_____ blood, will form _____ antibodies against the _____. Therefore, Rh group should also be matched before _____.	-ve, +ve, specific, Rh antigens, transfusions
44.	A special case of Rh incompatibility (mismatching) has been observed between the Rh _____ blood of a pregnant mother with Rh _____ blood of the foetus. _____ of the foetus do not get exposed to the Rh _____ blood of the mother in the first pregnancy as the two bloods are well separated by the placenta.	–ve, +ve, Rh antigens, –ve
45.	However, during the delivery of the first child, there is a possibility of exposure of the _____ blood to small amounts of the Rh+ve blood from the _____. In such cases, the _____ starts preparing _____ against Rh _____ in her blood.	Maternal, foetus, mother, antibodies, antigen
46.	In case of her subsequent pregnancies, the _____antibodies from the mother (Rh-ve) can leak into the blood of the foetus (Rh+ve) and destroy the _____.	Rh, foetal RBCs
47.	This could be fatal to the foetus or could cause severe anaemia and jaundice to the baby. This condition is called _____.	erythroblastosis foetalis
48.	This can be avoided by administering _____antibodies to the mother immediately after the delivery of the first child.	anti-Rh

	COAGULATION OF BLOOD	
49.	Blood exhibits _____ or clotting in response to an injury or trauma.	coagulation
50.	This is a mechanism to prevent _____ from the body.	excessive loss of blood
51.	You would have observed a _____ formed at the site of a cut or an injury over a period of time.	dark reddish brown scum
52.	It is a clot or _____ formed mainly of a network of threads called _____ in which _____are trapped.	coagulam, fibrins, dead and damaged formed elements of blood
53.	Fibrins are formed by the conversion of _____in the plasma by the enzyme _____.	inactive fibrinogens, thrombin
54.	Thrombins, in turn are formed from another inactive substance present in the plasma called _____.	prothrombin
55.	An enzyme complex, _____, is required for the above reaction.	thrombokinase
56.	This complex is formed by a series of linked enzymic reactions (_____) involving a number of factors present in the plasma in an _____ state.	cascade process, inactive
57.	An _____stimulates the _____ in the blood to release certain factors which activate the _____coagulation.	injury or a trauma, platelets, mechanism
58.	Certain factors released by the tissues at the site of injury also can initiate _____.	coagulation
59.	_____ions play a very important role in clotting.	Calcium
	LYMPH (TISSUE FLUID)	
60.	As the blood passes through the _____ in tissues, some water along with many small _____substances move out into the spaces between the cells of tissues leaving the _____and most of the formed _____in the blood vessels.	Capillaries, water soluble, larger proteins, elements
61.	This fluid released out is called the _____or _____. It has the same _____distribution as that in plasma.	interstitial fluid. tissue fluid, mineral
62.	Exchange of _____, gases, etc., between the blood and the cells always occur through this fluid.	nutrients
63.	An elaborate network of vessels called the _____collects this fluid and drains it back to the major veins. The fluid present in the lymphatic system is called the _____.	lymphatic system, lymph
64.	Lymph is a _____ fluid containing specialised _____ which are responsible for the _____ of the body. Lymph is also an important carrier for _____, etc.	colourless, lymphocytes, immune responses, nutrients, hormones
65.	_____ are absorbed through _____ in the _____ present in the intestinal villi.	Fats, lymph, lacteals

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	CIRCULATORY PATHWAYS	
66.	The circulatory patterns are of two types _____ or _____.	Open, closed
67.	Open circulatory system is present in _____ and _____ in which blood pumped by the heart passes through large vessels into open spaces or body cavities called _____.	Arthropods, molluscs, sinuses
68.	_____ and _____ have a closed circulatory system in which the blood pumped by the heart is _____ circulated through _____of blood vessels.	Annelids, chordates, always, a closed network
69.	This pattern is considered to be more _____ as the flow of fluid can be _____regulated.	Advantageous, more precisely
70.	All vertebrates possess a _____ chambered heart.	muscular
71.	_____ have a 2-chambered heart with an atrium and a ventricle.	Fishes
72.	_____ and the reptiles (except _____) have a 3-chambered heart with two atria and a single ventricle, whereas _____ and _____ possess a 4-chambered heart with two atria and two ventricles.	Amphibians, crocodiles, crocodiles, birds, mammals
73.	In fishes the heart pumps out _____ which is _____ by the _____ and supplied to the body parts from where deoxygenated blood is returned to the heart (_____ circulation).	Deoxygenated blood, oxygenated, gills, single
74.	In amphibians and reptiles, the left atrium receives oxygenated blood from the _____/_____/_____ and the right atrium gets the deoxygenated blood from other body parts. However, they get mixed up in the _____ _____ which pumps out mixed blood (_____ circulation).	Gills, lungs, skin, single ventricle, incomplete double
75.	In birds and mammals, oxygenated and deoxygenated blood received by the _____ and _____ atria respectively passes on to the ventricles of the same sides.	Left, right
76.	The ventricles pump it out _____ mixing up, i.e., two separate circulatory pathways are present in these organisms, hence, these animals have _____ circulation.	Without any, double
	HUMAN CIRCULATORY SYSTEM	
77.	Human circulatory system, also called the _____ _____ system consists of a _____ chambered heart, a network of closed branching blood vessels and _____, the fluid which is circulated.	Blood vascular, muscular, blood
78.	Heart, the _____ derived organ, is situated in the thoracic cavity, in between the two lungs, _____ to the left.	Mesodermally, slightly tilted
79.	It has the size of a _____. It is protected by a _____ walled _____ bag, _____, enclosing the pericardial fluid.	Clenched fist, double, membranous, pericardium

80.	Our heart has four chambers, two relatively _____ chambers called atria and two _____ chambers called ventricles.	Small upper, larger lower
81.	A _____,_____ wall called the _____ separates the right and the left atria, whereas a _____, the _____septum, separates the left and the right ventricles.	Thin, muscular, interatrial septum, thick-walled, inter-ventricular
82.	The atrium and the ventricle of the same side are also separated by a _____called the _____septum.	thick fibrous tissue, atrio-ventricular
83.	However, each of these septa are provided with _____through which the _____of the same side are connected.	an opening, two chambers
84.	The opening between the right atrium and the right ventricle is guarded by a _____ formed of _____or _____, the tricuspid valve, whereas a bicuspid or _____guards the opening between the left atrium and the left ventricle.	valve, three muscular flaps, cusps, mitral valve
85.	The openings of the right and the left ventricles into the pulmonary artery and the aorta respectively are provided with the _____.	semilunar valves
86.	The valves in the heart allows the _____only in _____direction, i.e., from the _____to the _____ and from the _____ to the pulmonary _____ or _____. These valves prevent any _____.	flow of blood, one, atria, ventricles, ventricles, artery, aorta, backward flow
87.	The entire heart is made of _____ muscles.	cardiac
88.	The walls of ventricles are much _____ than that of the atria.	thicker
89.	A _____ cardiac musculature called the _____is also distributed in the heart.	specialized, nodal tissue
90.	A patch of this tissue is present in the _____of the right atrium called the _____. Another mass of this tissue is seen in the _____corner of the right atrium close to the atrio-ventricular septum called the _____.	right upper corner, sino-atrial node (SAN), lower left, atrio-ventricular node (AVN)
91.	A bundle of _____, _____continues from the AVN which passes through the _____ septa to emerge on the _____ of the interventricular septum and immediately divides into a _____. These branches give rise to _____throughout the _____ musculature of the respective sides and are _____fibres.	nodal fibres, atrioventricular bundle (AV bundle), atrio-ventricular, top, right and left bundle, minute fibres, ventricular, called purkinje
92.	The nodal musculature has the _____without any external stimuli, i.e., it is _____.	ability to generate action potentials, autoexcitable

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93.	However, the number of _____that could be generated in a minute vary at different parts of the _____.	action potentials, nodal system
94.	The _____ can generate the _____ number of action potentials, i.e., _____min^{-1}, and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.	SAN, maximum, 70-75
95.	Therefore, it is called the _____.	pacemaker
96.	Our heart normally beats _____times in a minute (average _____ beats min^{-1}).	70-75, 72
	CARDIAC CYCLE	
97.	To begin with, _____ the four chambers of heart are in a _____ state, i.e., they are in _____.	All, relaxed, joint diastole
98.	As the tricuspid and bicuspid valves are open, blood from the _____ _____ and _____ cava flows into the left and the right ventricle respectively through the _____.	Pulmonary, veins, vena, left and right atria
99.	The semilunar valves are _____ at this stage.	closed
100.	The SAN now generates an _____which stimulates both the atria to undergo a _____ the atrial systole.	action potential, simultaneous, contraction
101.	This increases the _____into the ventricles by about _____ per cent.	flow of blood, 30
102.	The action potential is conducted to the _____ by the AVN and AV bundle from where the _____ transmits it through the entire _____ musculature.	ventricular side, bundle of His, ventricular
103.	This causes the ventricular muscles to contract, (_____), the atria undergoes _____, coinciding with the ventricular _____.	ventricular systole, relaxation (diastole), systole
104.	Ventricular systole increases the _____causing the closure of tricuspid and bicuspid valves due to attempted _____ of blood into the atria.	ventricular pressure, backflow
105.	As the _____increases further, the _____ guarding the pulmonary artery (right side) and the aorta (left side) are _____, allowing the blood in the ventricles to flow through these vessels into the circulatory pathways.	ventricular pressure, semilunar valves, forced open
106.	The ventricles now relax (ventricular diastole) and the ventricular pressure _____ causing the _____ of semilunar valves which prevents the backflow of blood into the ventricles.	falls, closure
107.	As the ventricular pressure declines further, the tricuspid and bicuspid valves are _____ by the pressure in the atria exerted by the blood which was being emptied into them by the _____.	pushed open, veins
108.	The blood now once again moves _____ to the ventricles.	freely

109.	The ventricles and atria are now again in a relaxed (_____) state, as earlier.	joint diastole
110.	This _____ event in the heart which is cyclically repeated is called the _____ and it consists of _____ and _____ of both the atria and ventricles.	sequential, cardiac cycle, systole, diastole
111.	As mentioned earlier, the heart beats _____ times per minute, i.e., that many_____ are performed per minute.	72, cardiac cycles
112.	From this it could be deduced that the duration of a _____ is 0.8 seconds.	Cardiac cycle
113.	During a cardiac cycle, each ventricle pumps out approximately _____ of blood which is called the _____ volume.	70 mL, stroke
114.	The stroke volume multiplied by the _____(no. of beats per min.) gives the _____.	cardiac output, heart rate
115.	Therefore, the _____can be defined as the volume of blood pumped out by each ventricle per minute and averages _____ mL or _____ litres in a healthy individual.	cardiac output, 5000, 5
116.	The body has the ability to alter the _____ as well as the _____and thereby the _____.	Stroke volume, heart rate, cardiac output
117.	For example, the _____of an athlete will be much _____ than that of an ordinary man.	cardiac output, higher
118.	During each cardiac cycle two prominent sounds are produced which can be easily heard through a _____.	stethoscope
119.	The first heart sound (_____) is associated with the closure of the _____valves whereas the second heart sound (_____) is associated with the closure of the _____ valves.	lub, tricuspid and bicuspid, dub, semilunar
	ELECTROCARDIOGRAPH (ECG)	
120.	This type of machine (_____) is used to obtain an _____(ECG).	electro-cardiograph, electrocardiogram
121.	ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle. A patient is connected to the machine with _____ electrical leads (one to each _____ and to the _____) that continuously monitor the heart activity.	three, wrist, left ankle
122.	For a detailed evaluation of the heart's function, _____are attached to the chest region.	multiple leads
123.	Each peak in the ECG is identified with a letter from P to T that corresponds to a specific _____of the heart.	electrical activity
124.	The P-wave represents the electrical excitation (or depolarisation) of the _____, which leads to the _____.	atria, contraction of both the atria
125.	The QRS complex represents the depolarisation of the _____, which initiates the _____.	ventricles, ventricular contraction

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126.	The _____ starts shortly after Q and marks the beginning of the _____.	contraction, systole
127.	The T-wave represents the return of the ventricles from _____state (_____).	excited to normal, repolarisation
128.	The end of the T-wave marks the end of _____.	systole
129.	Obviously, by counting the number of _____that occur in a given time period, one can determine the heart beat rate of an individual.	QRS complexes
130.	Since the ECGs obtained from different individuals have roughly the same shape for a given lead configuration, any deviation from this shape indicates a possible _____ or _____. Hence, it is of a great clinical significance.	Abnormality, disease
	DOUBLE CIRCULATION	
131.	The blood flows strictly by a fixed route through Blood Vessels the _____ and _____.	Arteries, veins
132.	Basically, each artery and vein consists of three layers: an inner lining of _____, the tunica _____, a middle layer of _____ and elastic fibres, the tunica _____, and an external layer of _____ tissue with _____ fibres, the tunica _____.	Squamous endothelium, intima, smooth muscle, media, fibrous connective, collagen, externa
133.	The tunica media is comparatively thin in the _____.	veins
134.	The blood pumped by the _____ enters the pulmonary artery, whereas the _____ pumps blood into the aorta.	right ventricle, left ventricle
135.	The _____ blood pumped into the _____ is passed on to the lungs from where the oxygenated blood is carried by the _____ into the left atrium.	deoxygenated, pulmonary artery, pulmonary veins
136.	This pathway constitutes the _____.	pulmonary circulation
137.	The oxygenated blood entering the _____ is carried by a network of arteries, arterioles and capillaries to the tissues from where the deoxygenated blood is collected by a system of _____, _____ and _____ and emptied into the right atrium. This is the _____ circulation.	aorta, venules, veins, vena cava, systemic
138.	The systemic circulation provides _____, _____ and other essential substances to the tissues and takes CO_2 and other harmful substances away for _____.	nutrients, O_2, elimination
139.	A _____ connection exists between the digestive tract and liver called _____.	unique vascular, hepatic portal system

140.	The _____ vein carries blood from intestine to the liver before it is delivered to the _____.	hepatic portal, systemic circulation
141.	A special _____ of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature.	coronary system
	REGULATION OF CARDIAC ACTIVITY	
142.	Normal activities of the heart are regulated _____, i.e., auto regulated by specialised muscles _____, hence the heart is called _____.	intrinsically, (nodal tissue), myogenic
143.	A special neural centre in the _____ can moderate the cardiac function through _____. Neural signals through the _____ nerves (part of _____) can increase the rate of heart beat, the strength of _____ and thereby the _____.	medulla oblangata, autonomic nervous system (ANS), sympathetic, ANS, ventricular contraction, cardiac output
144.	On the other hand, _____ neural signals (another component of _____) decrease the rate of heart beat, speed of conduction of action potential and thereby the _____.	parasympathetic, ANS, cardiac output
145.	Adrenal medullary hormones can also _____ the cardiac output.	increase
	DISORDERS OF CIRCULATORY SYSTEM	
146.	_____ is the term for blood pressure that is higher than normal (120/80).	Hypertension
147.	In this measurement 120 mm Hg (millimetres of mercury pressure) is the systolic, or _____, pressure and 80 mm Hg is the diastolic, or _____, pressure.	Pumping, resting
148.	If repeated checks of blood pressure of an individual is _____ (_____ over _____) or higher, it shows hypertension.	140/90, 140, 90
149.	High blood pressure leads to _____ diseases and also affects vital organs like brain and _____.	Heart, kidney
150.	Coronary Artery Disease, often referred to as _____, affects the vessels that supply blood to the _____.	Atherosclerosis, heart muscle
151.	It is caused by deposits of _____ and _____, which makes the lumen of arteries narrower.	calcium, fat, cholesterol, fibrous tissues
152.	Angina is also called _____. A symptom of _____appears when _____ is reaching the _____.	'angina pectoris', acute chest pain, no enough oxygen, heart muscle

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153.	Angina can occur in men and women of any age but it is more common among the _____and _____. It occurs due to conditions that affect the _____ flow.	middle-aged, elderly, blood
154.	_____ means the state of heart when it is not _____enough to meet _____.	Heart failure, pumping blood effectively, the needs of the body
155.	Heart failure is sometimes called _____because congestion of the lungs is one of the main symptoms of this disease.	congestive heart failure
156.	Heart failure is not the same as _____(when the heart stops beating) or a _____(when the heart muscle is suddenly damaged by an inadequate blood supply).	cardiac arrest, heart attack
	SUMMARY	
157.	_____ circulate blood, a fluid connective tissue, in their body, to transport essential substances to the cells and to carry waste substances from there.	Vertebrates
158.	Another fluid, _____ (tissue fluid) is also used for the transport of certain substances.	lymph
159.	Blood comprises of a fluid matrix, _____ and _____.	plasma, formed elements
160.	Red blood cells (RBCs, erythrocytes), white blood cells (WBCs, leucocytes) and platelets (_____) constitute the formed elements.	thrombocytes
161.	Blood of humans are grouped into A, B, AB and O systems based on the presence or absence of two surface antigens, A, B _____.	on the RBCs
162.	Another blood grouping is also done based on the presence or absence of another antigen called _____ on the surface of RBCs.	Rhesus factor (Rh)
163.	The spaces between cells in the tissues contain a fluid derived from blood called _____.	tissue fluid
164.	This fluid called _____ is almost similar to blood except for the _____ and the formed elements.	lymph, protein content
165.	All _____ and a _____have a closed circulatory system.	vertebrates, few invertebrates
166.	Our circulatory system consists of a _____, heart, a _____ and a fluid, blood.	muscular pumping organ, network of vessels
167.	Heart has two atria and two ventricles. Cardiac musculature is _____.	auto excitable

168.	Sino-atrial node (SAN) generates the _____ number of action protentials per minute (_____/min) and therefore, it sets the _____ of the activities of the heart. Hence it is called the _____.	maximum, 70-75, pace, Pacemaker
169.	The action potential causes the atria and then the ventricles to undergo _____ followed by their _____.	contraction (systole), relaxation (diastole)
170.	The _____ forces the blood to move from the atria to the ventricles and to the _____ artery and the aorta.	systole, pulmonary
171.	The cardiac cycle is formed by sequential events in the heart which is _____ and is called the cardiac cycle.	cyclically repeated
172.	A healthy person shows 72 such cycles _____.	per minute
173.	About 70 mL of blood is pumped out by each ventricle during a cardiac cycle and it is called the stroke or _____ volume.	beat
174.	Volume of blood pumped out by each ventricle of heart per _____ is called the cardiac output and it is equal to the product of _____ _____ and heart rate (approx 5 litres).	minute, stroke volume
175.	The _____ _____ of the heart can be recorded from the body surface by using _____ and the recording is called _____ (ECG) which is of clinical importance.	electrical activity, electrocardiograph, electrocardiogram
176.	We have a complete double circulation, i.e., two circulatory pathways, namely, _____ and _____ are present.	pulmonary, systemic
177.	The pulmonary circulation starts by the pumping of _____ by the right ventricle which is carried to the _____ where it is _____ and returned to the left atrium.	deoxygenated blood, lungs, oxygenated
178.	The _____ circulation starts with the pumping of _____ blood by the left ventricle to the _____ which is carried to all the body tissues and the _____ blood from there is collected by the _____ and returned to the _____ atrium.	systemic, oxygenated, aorta, deoxygenated, veins, right
179.	Though the heart is _____, its functions can be moderated by _____ and _____ mechanisms.	autoexcitable, neural, hormonal

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Breathing and Exchange of Gases

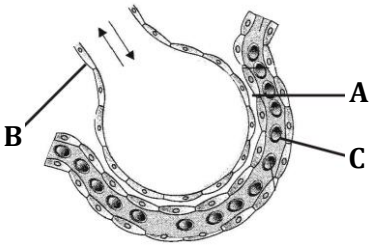
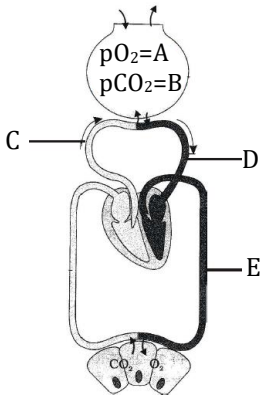
<table>
<tr><th>Q.NO.</th><th>QUESTIONS</th><th>ANSWERS</th></tr>
<tr><td>1.</td><td>Oxygen is utilised by the organism to ___(a)___ breakdown nutrients molecules like ___(b)___ and to derive ___(c)___ for performing various activities.</td><td>(a) Indirectly
(b) Glucose
(c) Energy</td></tr>
<tr><td>2.</td><td>Gas which is released during catabolic reaction of respiration is _____.</td><td>CO_2</td></tr>
<tr><td>3.</td><td>Process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is called</td><td>Breathing</td></tr>
<tr><td>4.</td><td>Mechanism of breathing vary among different groups of animals depending mainly on ____ and _____.</td><td>a. Habitats
b. Levels of organisation</td></tr>
<tr><td>5.</td><td>Match column-I and column-II
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>(A)</td><td>Diffusion over their entire body surface</td><td>(i)</td><td>Reptilia, Birds, mammals</td></tr>
<tr><td>(B)</td><td>Gills</td><td>(ii)</td><td>Insects</td></tr>
<tr><td>(C)</td><td>Moist cuticle</td><td>(iii)</td><td>Coelenterates, flat worm</td></tr>
<tr><td>(D)</td><td>Tracheal tubes</td><td>(iv)</td><td>Aquatic arthopods and molluscs</td></tr>
<tr><td>(E)</td><td>Vascularised bag</td><td>(v)</td><td>Earthworm</td></tr>
</table></td><td>A-iii, B-iv, C-v, D-ii, E-i</td></tr>
<tr><td>6.</td><td>Common passage for food and air_____.</td><td>Pharynx</td></tr>
<tr><td>7.</td><td>Glottis is the opening of _____.</td><td>Larynx</td></tr>
<tr><td>8.</td><td>Structure which in sound production is made up of _____.</td><td>Cartilages</td></tr>
<tr><td>9.</td><td>During swallowing glottis can be covered by a thick fibrous cartilaginous flap called epiglottis. (True/false)</td><td>False</td></tr>
<tr><td>10.</td><td>Origin of right and left primary bronchi occurs at the level of _____.</td><td>T_5</td></tr>
<tr><td>11.</td><td>Each terminal bronchiole gives rise to a number of very thin, irregular walled and vascularised bag like structure called _____.</td><td>Alveoli</td></tr>
<tr><td>12.</td><td>Incomplete cartilaginous rings are present upto _____.</td><td>Initial Bronchioles</td></tr>
<tr><td>13.</td><td>Conducting part starts with external nostrils and extended upto_____.</td><td>Terminal bronchiole</td></tr>
<tr><td>14.</td><td>What is the significance of pleural fluid ?</td><td>Provide friction-less movement between pleural membrane</td></tr>
<tr><td>15.</td><td>Given statement is true or false.
Statement : Actual site for diffusion of O_2 and CO_2 between blood and atmospheric air is exchange part.</td><td>True</td></tr>
</table>

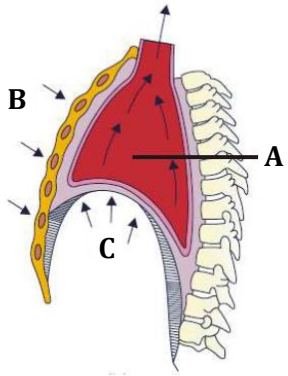
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<table>
<tr><td>16.</td><td>Breathing involves two stages ___(a)____ during which atmospheric air is drawn in and __(b)__ by which the alveolar air is released out.</td><td>a. Inspiration
b. Expiration</td></tr>
<tr><td>17.</td><td>Movement of air into and out of the lungs is carried out by a creating ____ between the lungs and atmosphere.</td><td>Pressure gradient</td></tr>
<tr><td>18.</td><td>Expiration takes place when</td><td>Intrapulmonary pressure is higher than the atmospheric pressure</td></tr>
<tr><td>19.</td><td>The volume of air involved in breathing movements can be estimated by :</td><td>Spirometer</td></tr>
<tr><td>20.</td><td>Match column-I and column-II.
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>(a)</td><td>Tidal volume</td><td>(i)</td><td>1200 ml</td></tr>
<tr><td>(b)</td><td>Expiratory reserve volume</td><td>(ii)</td><td>4200 ml</td></tr>
<tr><td>(c)</td><td>Inspiratory capacity</td><td>(iii)</td><td>500 ml</td></tr>
<tr><td>(d)</td><td>Vital capacity</td><td>(iv)</td><td>1100 ml</td></tr>
<tr><td>(e)</td><td>Residual volume</td><td>(v)</td><td>2300 ml</td></tr>
<tr><td>(f)</td><td>Functional residual capacity</td><td>(vi)</td><td>3000 ml</td></tr>
</table></td><td>a-iii, b-iv, c-vi, d-ii, e-i, f-v</td></tr>
<tr><td>21.</td><td>Simple diffusion mainly based on :</td><td>(a) Pressure/concentration Gradient
(b) Solubility of gases
(c) Thickness of membrane involved in diffusion</td></tr>
<tr><td>22.</td><td>The diffusion membrane is made up of :
(A) Thick columnar epithelium of alveoli
(B) Endothelium of alveolar capillaries
(C) Basement substances in between capillaries and alveoli</td><td>Only (B and C)</td></tr>
<tr><td>23.</td><td>Least amount of CO_2 is carried out in _____</td><td>Dissolved state through plasma</td></tr>
<tr><td>24.</td><td>Factors those interfere with the binding of O_2 with haemoglobin are :</td><td>(A) Partial pressure of CO_2
(B) H^+ concentration,
(C) Temperature</td></tr>
<tr><td>25.</td><td>O_2 dissociation curve is :</td><td>Sigmoid curve</td></tr>
<tr><td>26.</td><td>Factor which is favourable for the formation of oxyhaemoglobin.</td><td>(A) Lesser H^+ concentration
(B) High pO_2, (C) Low pCO_2</td></tr>
<tr><td>27.</td><td>Under normal physiological condition every 100 ml of oxygenated blood can deliver ___ml of O_2.</td><td>5</td></tr>
<tr><td>28.</td><td>Dissociation of CO_2 from carbamino-haemoglobin takes place when :</td><td>pCO_2 is low ; pO_2 is high</td></tr>
<tr><td>29.</td><td>At the tissue site where partial pressure of CO_2 is _a_ due to _b_, CO_2 diffuse into blood and form _c_ and _d_.</td><td>a - High, b - Catabolism
c - HNO_3^-, d - H^+</td></tr>
</table>

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<table>
<tr><td>30.</td><td>Every 100 mL of deoxygenated blood transport how much amount of CO_2 ?</td><td>4 ml</td></tr>
<tr><td>31.</td><td>Human beings have ability to maintain and moderate the respiratory rhythm by the help of :</td><td>Neural system</td></tr>
<tr><td>32.</td><td>Respiratory rhythm centre is present in _____ region of brain</td><td>Medulla oblongata</td></tr>
<tr><td>33.</td><td>Centre which can moderate the function of respiratory rhythm centre is located in :</td><td>Pons</td></tr>
<tr><td>34.</td><td>Chemosensitive area sends signals to make necessary adjustments for respiration to :</td><td>Respiratory rhythm centre</td></tr>
<tr><td>35.</td><td>Aortic arch and carotid artery receptors can recognize changes in :</td><td>CO_2, H^+</td></tr>
<tr><td>36.</td><td>Role of oxygen in the regulation of respiratory rhythm is quite insignificant. (True/false)</td><td>True</td></tr>
<tr><td>37.</td><td>Match column-I and column-II.
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>(A)</td><td>Asthma</td><td>(i)</td><td>Alveolar wall damaged</td></tr>
<tr><td>(B)</td><td>Emphysema</td><td>(ii)</td><td>Inflammation leading to fibrosis</td></tr>
<tr><td>(C)</td><td>Occupational respiratory disorder</td><td>(iii)</td><td>Wheezing due to inflammation of bronchi and bronchioles</td></tr>
</table></td><td>A-iii, B-i, C-ii</td></tr>
<tr><td>38.</td><td>Long exposure to dust can give rise to :</td><td>Inflammation leading to proliferation of fibrous tissues in lungs.</td></tr>
<tr><td>39.</td><td>Identify A, B and C in given diagram :
 
</td><td>(A) Basement substance
(B) Alveolar wall
(C) RBC</td></tr>
<tr><td>40.</td><td>Identify A, B, C, D, E in given diagram :
 
</td><td>(A) 104
(B) 40
(C) Pulmonary artery
(D) Pulmonary vein
(E) Systemic arteries</td></tr>
</table>

41.	Identify A, B, and C in the given diagram :  	(A) Volume of thorax decreased (B) Ribs and sternum return to original position (C) Diaphragm relaxed

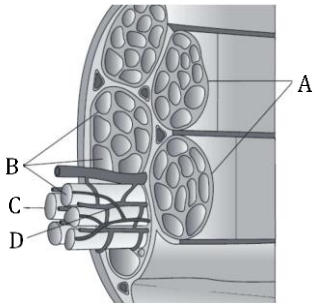
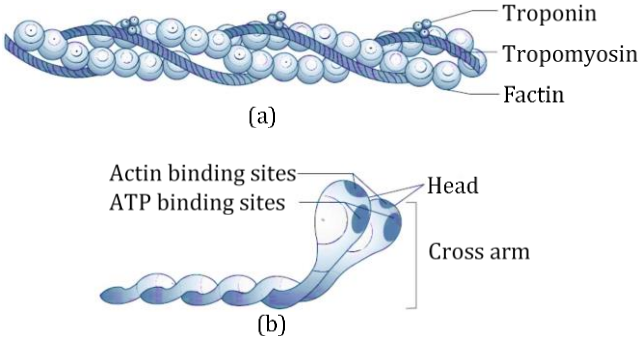
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Locomotion and Movement
(Muscles)

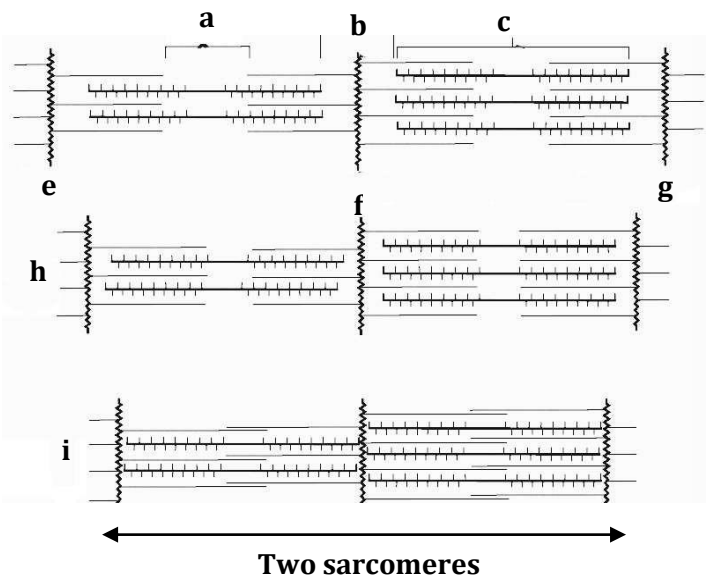
Q.NO.	QUESTIONS	ANSWERS
1.	Muscle is a specialised tissue of _____ origin.	MESODERMAL
2.	Muscles have special properties like ___, ____, ____and ____.	Excitability, contractibility, elasticity
3.	Three types of muscles are	Skeletal, visceral, cardiac
4.	_____muscles are closely associated with the skeletal components of the body.	Skeletal
5.	Skeletal muscles have a striped appearance under the microscope and hence called _____muscles.	Striated
6.	Name the muscles which are closely related with the locomotory actions and changes in body postures.	Skeletal
7.	__A__ muscles are located in the inner walls of hollow visceral organs of the body like the alimentary canal, reproductive tract etc. They do not exhibit any __B__ and are in __C__ appearance.	(A) visceral (B) striation (C) Smooth
8.	Smooth muscles are also called involuntary muscles. Why ?	Because these are not under the voluntary control of nervous system
9.	Name the muscles which assist in the transportation of food through the digestive tract and gametes through the genital tract.	Smooth muscles (involuntary muscles)
10.	Based on appearance cardiac muscles are _____.	Striated
11.	Each organised skeletal muscle in our body is made of a number of _____ held together by a common collagenous connective tissue layer called _____.	Muscle bundles or fascicles, fascia
12.	What is sarcolemma?	Plasma membrane of muscle fibre is called sarcolemma
13.	Store house of Ca^{+2} in muscle fibre is ____	Endoplasmic reticulum
14.	A characteristic feature of a muscle fibre is the presence of large number of parallel arranged filaments in the sarcoplasm called____.	Myofibrils or myofilaments
15.	Light bands of muscle fiber consist of filaments called ____whereas dark bands consist of filaments called____.	Actin, myosin
16.	Fill in the blanks with the suitable words _____ filaments are thinner as compared to the____ filaments, hence are commonly called _____ and ____ filaments respectively.	Actin, myosin, thin, thick
17.	In the center of each I band is an elastic fiber, called ____.	Z line

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18.	Fill in the blanks with the suitable words : The thick filaments in the 'A' band are also held together in the middle of this band by a thin fibrous membrane called __a__. The 'A' and 'I' bands are arranged alternately throughout the length of the __b__. The portion of the myofibril between two successive 'Z' lines is considered as the functional unit of contraction and is called a __c__.	(a) M line (b) myofibrils (c) sarcomere
19.	Fill in the blanks with the suitable words : Each actin (thin) filament is made of two 'F' (filamentous) actins __a__ wound to each other. Each 'F' actin is a polymer of __b__ actins. Two filaments of another protein __c__ also run close to the 'F' actins throughout its length. A complex protein __d__ is distributed at regular intervals on the tropomyosin. In the resting state a submit of troponin masks the active binding sites for __e__ on the __f__ filaments.	(a) Helically (b) monomeric G (globular) (c) tropomyosin (d) troponin (e) myosin (f) actin
20.	Name the two important parts of meromyosin.	Globular head with a short arm and a tail
21.	Label A, B, C and D in the given diagram.  	(A) fascicle (B) muscle fibre (C) sarcolemma (D) blood capillary
22.	Identify the given diagrams.  	(a) an actin filament (b) myosin monomer (meromyosin)
23.	Mechanism of muscle contraction is best explained by _____.	Sliding filament theory
24.	Give only the statement of sliding filament theory.	Contraction of a muscle fiber takes place by the sliding of the thin filaments over the thick filaments

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<table>
<tr><td>25.</td><td>Complete the labelling's of the following diagram.
 
</td><td>(a) H zone
(b) I band
(c) A band
(d) relaxed
(e) Z line
(f) Z line
(g) Z line
(h) contracting
(i) maximally contracted</td></tr>
<tr><td>26.</td><td>Store house of Calcium ions for muscle contraction is_____.</td><td>Sarcoplasmic reticulum</td></tr>
<tr><td>27.</td><td>What type of movement is seen in Amoeba ?</td><td>Streaming movement of protoplasm</td></tr>
<tr><td>28.</td><td>Cells of the human body exhibit three main types of movements _____, _____ and _____.</td><td>Amoeboid, ciliary, muscular</td></tr>
<tr><td>29.</td><td>About _____ percent of the body weight of a human adult is contributed by muscles.</td><td>40-50</td></tr>
<tr><td>30.</td><td>Match the column :-
<table>
<tr><th colspan="2">Column A</th><th colspan="2">Column B</th><th colspan="2">Column C</th></tr>
<tr><td>a.</td><td>Striped muscles</td><td>i.</td><td>Located in the inner walls of hollow visceral organs of the body</td><td>1</td><td>Involved in locomotory actions</td></tr>
<tr><td>b.</td><td>Visceral muscles</td><td>ii.</td><td>Muscles of heart</td><td>2</td><td>Changes of body posture</td></tr>
<tr><td>c.</td><td>Cardiac muscles</td><td>iii.</td><td>Closely associated with the skeletal components of the body</td><td>3</td><td>Transportation of food</td></tr>
<tr><td></td><td></td><td></td><td></td><td>4</td><td>Contraction and relaxation of heart</td></tr>
</table></td><td>a-iii-1, 2,
b-i-3,
c-ii-4</td></tr>
</table>

31.	Muscle fibre is a _____ as the sarcoplasm contains many nuclei.	Syncytium
32.	Name the two important proteins due to which striated appearance is seen in myofibril.	Actin, myosin
33.	Name the protein found in isotropic and anisotropic bands of myofibrils respectively ?	Actin, myosin
34.	The main source of energy during muscle contraction is _____.	Glycogen
35.	Dark bands of myofibrils are known as _____ band.	Anisotropic
36.	Muscle contraction is initiated by a signal sent by the _____.	Central nervous system
37.	The functional unit of the contractile system in the striped muscle is _____.	Sarcomere
38.	Contraction of a muscle is caused by -	Actomyosin
39.	During muscle contraction ___a___ energy is converted into ___b___ energy.	(a) chemical (b) mechanical
40.	The distance or length of the myofibril between two adjacent z-bands is called.	Sarcomere
41.	In a relaxed fibril, H-zone, a lighter region of low density can be seen in the centre of _____.	A-band

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Locomotion and Movement
(Skeletal System)

Q.NO.	QUESTIONS	ANSWERS
1.	Skeletal system consist of a framework of _____ and a few ____.	Bones, cartilages
2.	How many bones are found in axial skeleton?	80
3.	Axial skeleton consist of ____bones distributed _____.	80, along the main axis of the body
4.	The skull, vertebral column and ribs constitute the ____ skeleton.	Axial
5.	The skull consist of ___sets of bones ____ and ____.	Two, cranial, facial
6.	How many bones are found in human cranium?	8
7.	Name the U shaped bone present at the base of the skull.	Hyoid
8.	Each middle ear consist of three tiny bones - ___, ___and __ collectively called _____.	Malleus, incus, stapes, ear ossicles
9.	The skull region articulates with the superior region of the vertebral column with the help of two _____.	Occipital condyles
10.	Type of skull found in human beings is _____.	Dicondylic
11.	How many bones are found in human vertebral column ?	26
12.	First vertebrae is _____.	Atlas
13.	Name the central hollow portion in vertebrae through which the spinal cord passes.	Neural canal
14.	The number of cervical vertebrae are ____in almost all mammals including human beings.	Seven
15.	_____ is a flat bone on the midventral line of the thorax.	Sternum
16.	Write the functions of the vertebral column.	Protection of spinal cord, support the head, serves as point of attachment for the ribs and musculature of the back
17.	Ribs are dorsally attached with ___ whereas ventrally with ___.	Vertebral column, sternum
18.	Each forelimb consist of ___bones.	30
19.	The bones of the limbs along with their girdles constitute the ___.	Appendicular skeleton
20.	Longest and strongest bone of the human body is _____.	Femur
21.	A cup shaped bone called _____ cover the knee ventrally.	patella
22.	Each half of pectoral girdle consist of a ____ and a ___.	Clavicle, scapula

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<table>
<tr><td>23.</td><td>Name the type of joint between the following :-
(a) atlas/axis
(b) carpal/metacarpal of thumb
(c) between phalanges
(d) femur/acetabulum
(e) between cranial bones
(f) between pubic bones in the pelvic girdle</td><td>
(a) pivot
(b) saddle
(c) hinge
(d) ball and socket
(e) sutures
(f) pubis symphysis</td></tr>
<tr><td>24.</td><td>Fill in the blank spaces :
(a) All mammals (except a few) have _____ cervical vertebra.
(b) The number of phalanges in each limb of human a _____.
(c) _____ and _____ pairs of ribs are called floating ribs.
(d) The human cranium is made of _____ bones.</td><td>
(a) seven
(b) 14
(c) 11th, 12th
(d) 8</td></tr>
<tr><td>25.</td><td>Write true or false.
(a) Arthritis is inflammation of joints
(b) Osteoporosis is a age related disorder characterized by decreased bone mass and increased chances of fractures.
(c) Gout is the inflammation of joints due to accumulation of calcium carbonate crystals</td><td>
(a) true
(b) true

(c) false</td></tr>
<tr><td>26.</td><td>Name the age related disorder characterized by decreased bone mass and increased chances of fractures.</td><td>Osteoporosis</td></tr>
<tr><td>27.</td><td>In joints, the bones involved are joined together with the help of cartilages. The joint between the adjacent vertebrae in the vertebral column is of this pattern and it permits limited movements, is –</td><td>Cartilagenous</td></tr>
<tr><td>28.</td><td>_____ joints are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.</td><td>Synovial</td></tr>
<tr><td>29.</td><td>Match the following bones of vertebral column.
<table>
<tr><th></th><th>Name</th><th></th><th>Number</th></tr>
<tr><td>A.</td><td>CERVICAL</td><td>I.</td><td>7</td></tr>
<tr><td>B.</td><td>THORACIC</td><td>II.</td><td>12</td></tr>
<tr><td>C.</td><td>LUMBAR</td><td>III.</td><td>5</td></tr>
<tr><td>D.</td><td>SACRAL</td><td>IV.</td><td>(5)</td></tr>
<tr><td>E.</td><td>COCCYX</td><td>V.</td><td>(4)</td></tr>
</table></td><td>A–I, B–II, C–III,
D–IV, E–V</td></tr>
<tr><td>30.</td><td>Match the following columns.
<table>
<tr><th></th><th>Name</th><th></th><th>Number</th></tr>
<tr><td>A.</td><td>Vertebrosternal ribs</td><td>I.</td><td>14</td></tr>
<tr><td>B.</td><td>Vertebrochondral ribs</td><td>II.</td><td>4</td></tr>
<tr><td>C.</td><td>Floating ribs</td><td>III.</td><td>6</td></tr>
</table></td><td>A–I, B–III, C–II</td></tr>
<tr><td>31.</td><td>Match the following bones of forelimbs.
<table>
<tr><th></th><th>Name</th><th></th><th>Number</th></tr>
<tr><td>A.</td><td>Humerus</td><td>I.</td><td>14</td></tr>
<tr><td>B.</td><td>Carpals</td><td>II.</td><td>8</td></tr>
<tr><td>C.</td><td>Metacarpals</td><td>III.</td><td>5</td></tr>
<tr><td>D.</td><td>Phalanges</td><td>IV.</td><td>1</td></tr>
</table></td><td>A–IV, B–II, C–III, D–I</td></tr>
</table>

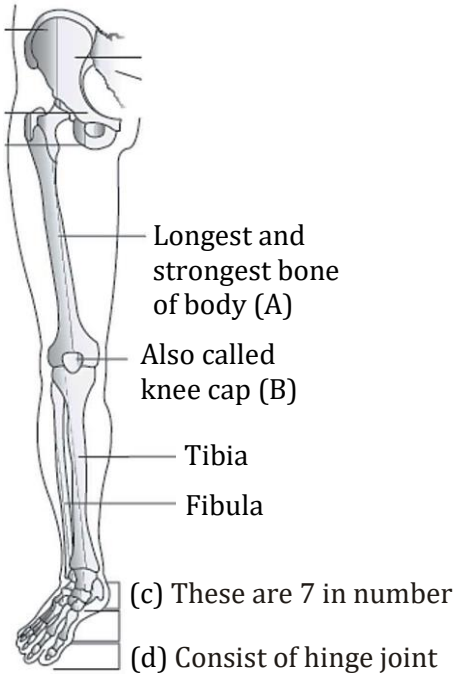
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32. Match the following bones of hindlimbs.

	Name		Number
A.	Femur	I.	14
B.	Tarsals	II.	7
C.	Metatarsals	III.	5
D.	Phalanges	IV.	1

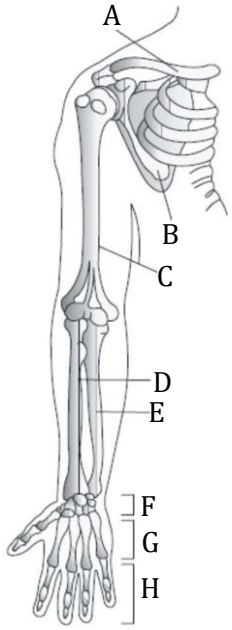
A–IV, B–II, C–III, D–I

33. Identify A , B , C and D in the given diagram.



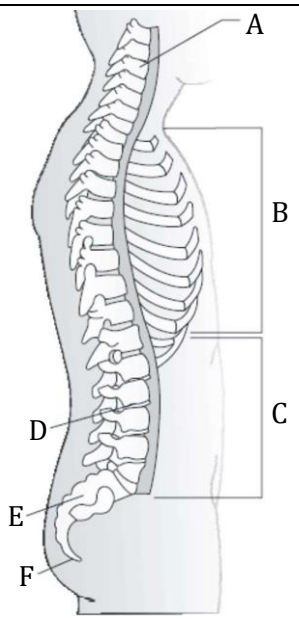
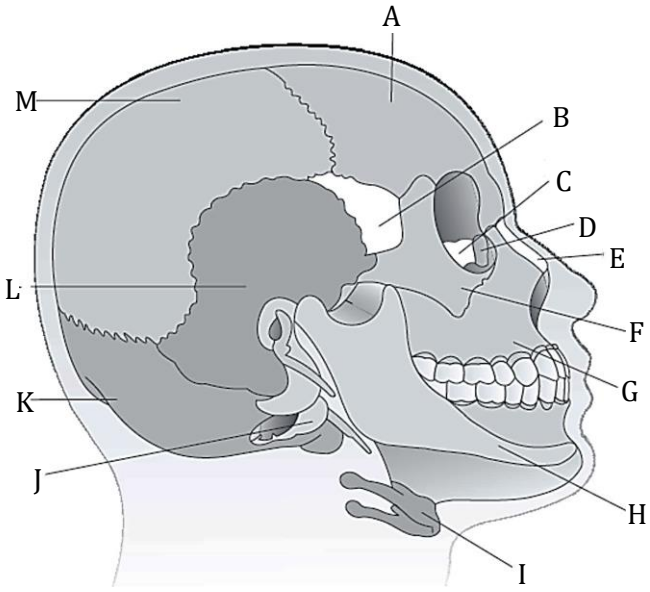
(A) Femur
(B) Patella
(C) Tarsals
(D) Phalanges

34. Label A , B , C , D , E , F , G and H in the given diagram of forelimb.



(A) clavicle
(B) scapula
(C) humerus
(D) radius
(E) ulna
(F) carpals
(G) metacarpals
(H) phalanges

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35.	 Identify the labelling A , B , C , D , E and F in the given diagram of vertebral column.	A. Cervical vertebrae B. Thoracic vertebrae C. Lumbar vertebrae D. Intervertebral disc E. Sacrum F. coccyx
36.	 Identify A , B , C , D , E , F , G , H , I , J , K , L and M in the given diagram of skull.	A. Frontal bone B. Sphenoid bone C. Ethmoid bone D. Lacrimal bone E. Nasal bone F. Zygomatic bone G. Maxilla H. Mandible I. Hyoid bone J. Occipital condyle K. Occipital bone L. Temporal bone M. Parietal bone

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Excretory Products and Their Elimination

Q.NO.	QUESTIONS	ANSWERS
1.	Common forms of nitrogenous waste of organism are	Ammonia, urea, uric acids.
2.	Most toxic nitrogenous waste material is	Ammonia.
3.	Least toxic nitrogenous waste material is	Uric acid.
4.	Removal of ammonia from body is called	Ammonotelism.
5.	Length of kidney in human being is	10-12 cm
6.	Average weight of human kidney in human beings is	120 to 170 gm
7.	The notch on the centre of inner concave surface of kidney is	Hilum
8.	Funnel shaped space in kidney is called	Renal pelvis
9.	Projections of renal pelvis called	Calyces
10.	Outer zone of kidney is known as	Cortex
11.	Inner zone of kidney is known as	Medulla
12.	Name the conical masses of medulla	Medullary pyramids
13.	Number of nephron in each kidney is	1 Million
14.	Name the branch of artery which take away blood from glomerulus	Efferent arterioles
15.	The structure which encloses glomerulus is	Bowman's Capsule
16.	Glomerulus and Bowman's capsule together form	Malpighian body
17.	Hair pin shaped structure in nephron is called	Henle's loop
18.	DCT of nephron opens in	Collecting duct
19.	Loop of Henle is too short and remain very little into medulla. This nephron is _____.	Cortical nephron
20.	Loop of Henle come deep into medulla in	Juxta medullary nephron
21.	Projection of cortex into medulla is called	Columns of Bertini
22.	U-shaped vessels running parallel to Henle's loop called	Vasa recta
23.	Vasa recta is absent in	Cortical nephron
24.	Epithelial cells of Bowman's capsule	Podocytes
25.	GFR is regulated by	Juxta glomerular apparatus
26.	Approximate value of GFR	180 litre/day
27.	A special sensitive region formed by cellular modifications in DCT and afferent arteriole at the location of their contact is called.	Juxta glomeular apparatus
28.	Reabsorption occurs in	Tubular epithelial cells
29.	PCT is lined by ..... epithelium	Simple cuboidal brush border

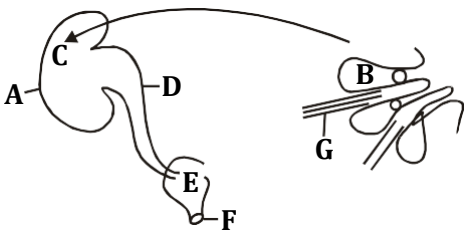
30.	How much electrolytes are absorbed by PCT of nephron ?	70 to 80%
31.	Flow of blood in opposite direction in limbs of henle's loop and vasa recta forms	Counter current mechanism
32.	Counter current mechanism is useful for	Maintaining concentration gradient in medullary interstitium
33.	pH of urine is	'6'
34.	On an average how much urea is excreted out per day ?	25-30 gm
35.	Presence of glucose in urine is called	Glycosuria
36.	Presence of ketone bodies in urine is called	Ketonuria
37.	Ketonuria & glycosuria is an indicative of	Diabetes mellitus
38.	Our lungs remove large amounts of CO_2	approximately 200 ml/min.
39.	Primary function of sweat is	Cooling of the body
40.	Accumulation of urea in blood is called	Uremia
41.	Stones or insoluble mass of crystallised salts formed within the kidney is called.	Renal calculi
42.	Inflammation of glomeruli of kidney is called	Glomerulonephritis

Q.NO.	FILL IN THE BLANKS	ANSWERS
1.	In kidney failure condition urea can be removed by _____ process.	Hemodialysis
2.	Ammonia is converted into urea in _____ and filtered from blood and excreted out by _____.	Liver, Kidney
3.	Kidney is situated between level of last _____ and third_____ vertebra.	Thoracic, lumbar
4.	Projection of cortex between medullary pyramids is called _____	Column of bertini
5.	_____ is a tuft of capillaries formed by the afferent arteriole which is a fine branch of _____.	Glomerulus, renal artery
6.	Efferent arteriole emerging from glomerulus forms fine capillary network around renal tubule known as _____ capillaries.	Peritubular
7.	_____, _____ and _____ are three processes involved in urine formation.	Glomerular filtration, reabsorption, secretion
8.	Epithelial cells of bowman's capsule called ____ are arranged in an _____ manner.	Podocytes Intricate
9.	A fall in GFR can activate the juxta glomerular cells to release _____.	Renin

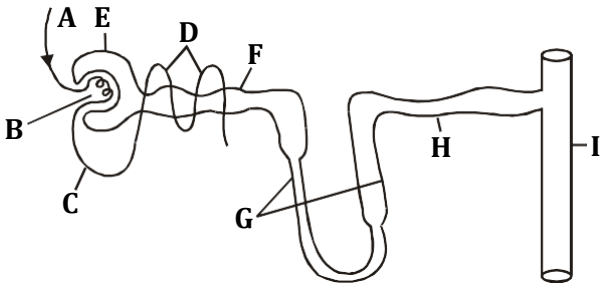
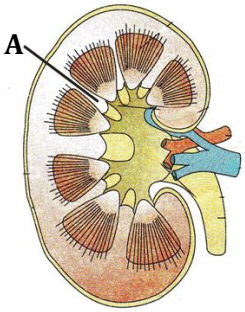
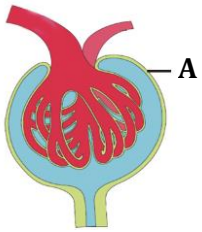
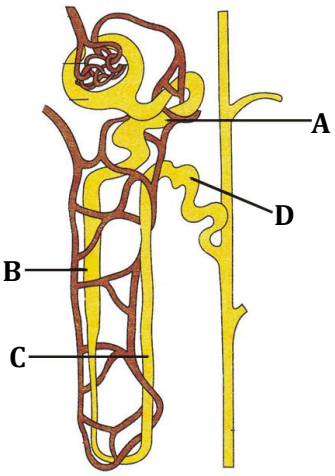
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10.	Glucose, amino acids and Na^+ are reabsorbed _____.	Actively
11.	Tubular cells secrete _____ and _____.	H^+, K^+
12.	Tubular secretion helps in maintenance of _____ and _____ balance of the body.	Ionic, acid-base
13.	PCT helps in secretion of _____.	H^+, K^+ and NH_3
14.	Descending limb of loop of Henle is _____ and ascending limb is _____ to water.	Permeable, impermeable
15.	Kidney function is regulated by feedback mechanism involving the_____ and _____.	Hypothalamus, juxta glomerulus apparatus
16.	Osmoreceptors in the body are activated by change in blood volume, body fluid volume and ___A___. An excessive___B___ of fluid from the body can activate these receptors which stimulate ___C___ to release ___D___ from the ___E___.	A - ionic concetration, B - loss, C - hypothalamus, D - ADH, E - neurohypophysis
17.	ANF causes _____.	Vasodilation
18.	Fall in GFR activate JG cells to release renin which converts angiotensinogen into _____. Angiotensinogen-II is a powerful _____. Causes increase in glomerular blood pressure. Causes increase in _____.	 Angiotensin-I Vasoconstrictor GFR
19.	Angiotensinogen-II also stimulates adrenal cortex to release _____.	Aldosterone
20.	**(i)** Signals from CNS started by stretching of the _____ as it gets filled with urine. **(ii)** Stretch receptors send signals to _____. **(iii)** CNS gives message to motor nerves to initiate contraction in ____ and simultaneous relaxation of the ____ causing release of urine.	Urinary bladder CNS Smooth muscles of bladder, urethral Sphincter

Q.NO.	DIAGRAM BASED QUESTIONS	ANSWERS
1.	Identify the labellings in the given diagram.  	A. Renal cortex B. Medullary pyramid C. Renal pelvis D. Ureter E. Urinary bladder F. Urethra G. Renal columns of Bertini

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2.		A. Afferent arteriole B. Glomerulus C. Efferent arteriole D. Peritubular capillaries E. Bowman's capsule F. Proximal convoluted tubule G. Loop of Henle H. Distal convoluted tubule I. Collecting duct
3.	Labelled part A is the part of :- 	Cortex
4.	Which type of tissue is covered the labelling part. 	Simple squamous
5.	Identify the marked part with their tissue. 	A. Proximal convoluted tubule-Simple Cuboidal brush border epithelium B. Descending limb of loop of Henle - simple squamous C. Ascending limb of loop of Henle (Thin) - simple squamous epithelium D. Distal convoluted tubule - simple cuboidal epithelium

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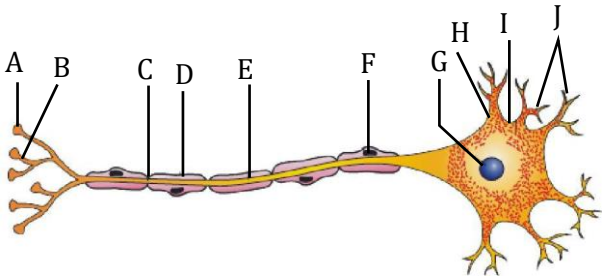
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<tr><th>Q.NO.</th><th>MATCH THE COLUMN-I WITH COLUMN-II</th><th>ANSWERS</th></tr>
<tr><td>1.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>(i)</td><td>Ammonotelic</td><td>(a)</td><td>Aquatic insects</td></tr>
<tr><td></td><td></td><td>(b)</td><td>Amphibians</td></tr>
<tr><td></td><td></td><td>(c)</td><td>Bony fishes</td></tr>
<tr><td>(ii)</td><td>Ureotelic</td><td>(d)</td><td>Marine fishes</td></tr>
<tr><td></td><td></td><td>(e)</td><td>Aquatic amphibians</td></tr>
<tr><td></td><td></td><td>(f)</td><td>Reptiles</td></tr>
<tr><td>(iii)</td><td>Uricotelic</td><td>(g)</td><td>Birds</td></tr>
<tr><td></td><td></td><td>(h)</td><td>Land snails</td></tr>
</table>
</td><td>(i) a, c, e;
(ii) b, d;
(iii) f, g, h</td></tr>
<tr><td>2.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>(i)</td><td>Vertebrates</td><td>(a)</td><td>Flame cells</td></tr>
<tr><td>(ii)</td><td>Earthworm</td><td>(b)</td><td>Green glands</td></tr>
<tr><td>(iii)</td><td>Platyhelminthes</td><td>(c)</td><td>Malpighian tubules</td></tr>
<tr><td>(iv)</td><td>Crustaceans</td><td>(d)</td><td>Kidney</td></tr>
<tr><td>(v)</td><td>Cockroach</td><td>(e)</td><td>Nephridia</td></tr>
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</td><td>(i)-d,
(ii)-e,
(iii)-a,
(iv)-b,
(v)-c</td></tr>
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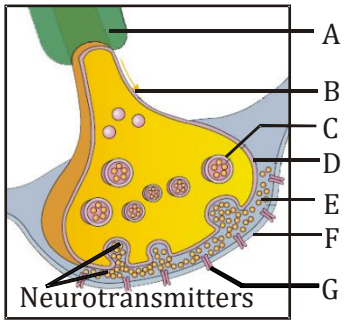
Neural Control and Coordination

Q.NO.	QUESTIONS	ANSWERS
1.	Process through two or more organ interact and compliment the function of each other is :-	Coordination
2.	Other system from which nervous system coordinate & integrate in	Endocrine system
	synchronised fashion is :-	
3.	Organised network of neural system, which provides quick coordination through.	Point to point connections
4.	Highly specialised cells in animals which can detect, receive & transmit different kinds of stimuli.	Neurons
5.	Lower vertebrate in which neural organisation is very simple & composed of network of neurons is	Hydra
6.	Site where all the information are processed & controlled is	CNS
7.	Part of PNS which carries impulses from tissue/organ to CNS are	Afferent fibres
8.	Neural system which relays impulses from CNS to skeletal muscle is	Somatic neural system
9.	Branch of PNS which transmit impulses from CNS to involuntary organ and classified into sympathetic & parasympathetic neural system is	(A.N.S) Autonomic neural system
10.	Bulb like structure where branch of nerve fibre terminates	Synaptic knob
11.	Chemicals by which nerve impulses transmit through one neuron to another are	Neurotransmitters
12.	Cells which envelop nerve fibres and form myelin sheath around the axon are	Schwann cells
13.	Gap between two adjacent myelin sheath are known as	Node of Ranvier
14.	Myelinated and unmyelinated nerve fibres are found in PNS respectively are	Cranial/Spinal nerve and Autonomic & somatic nerve fibres
15.	Property of excitability of neuron is due to	Polarised state of membrane of neuron
16.	At resting stage permeability of K+ Ion is more comparatively Na+ Ion through axonal membrane	True
17.	<table><tr><th colspan="2">Types of neuron</th><th colspan="2">Source</th></tr><tr><td>(a)</td><td>Multipolar</td><td>(i)</td><td>Embryonic stage</td></tr><tr><td>(b)</td><td>Bipolar</td><td>(ii)</td><td>Cerebral cortex</td></tr><tr><td>(c)</td><td>Unipolar</td><td>(iii)</td><td>Retina of eye</td></tr></table> (1) a-i, b-iii, c-ii (2) a-iii, b-ii, c-i (3) a-ii,b-iii, c-i (4) a-ii, b-i, c-iii	a-ii, b-iii, c-i

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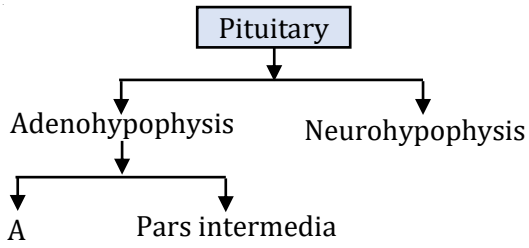
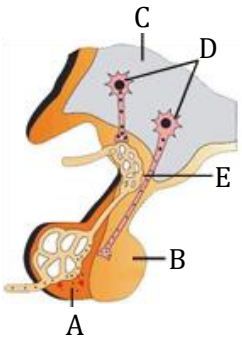
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18.	At resting stage charge of inner surface is due to presence of	More K^+ Ion and negatively charge protein molecule
19.	Concentration gradient of axoplasm at resting stage is maintained by	Na^+ K^+ pump
20.	When a stimulus is applied on axonal membrane then generation of action potential is due to	Change permeability for Na^+ Ion
21.	Generation of action potential in stage of	Exciting stage
22.	Junctions from which impulse is transmitted through one neuron to another are known as	Synapse
23.	Type of synapse in which transmission of impulse across neuron is similar to conduction along a single axon	Electrical synapse
24.	Inside the skull the brain is covered by some membrane are called as	Meninges/meninx
25.	Deep cleft which divides cerebrum longitudinally into two halves, known as	Median fissure
26.	Tract of nerve fibre which connects both cerebral hemisphere is	Corpus callosum
27.	The portion of cerebrum which is greyish in appearance is	Cerebral cortex
28.	Region of cerebral cortex which is neither motor nor sensory is	Association area
29.	Major coordinating centre of fore brain which is wrapped around by cerebrum and responsible for motor & sensory signalling is	Thalamus
30.	Important part of fore brain which control body temperature urge for eating & drinking and has several group of neurosecretory cells is	Hypothalamus
31.	Inner part of cerebral hemisphere, which is involved in sexual behaviour, motivation expression of emotional reactions etc. and a group of associated deep structure like amygdala, hippocampus etc. is	Limbic system
32.	Canal of mid brain which connects forebrain & hindbrain is	Cerebral aqueduct
33.	Round swelling at the dorsal portion of the midbrain is known as	Corpora Quadrigemina
34.	Given below is a structure of a neuron label the parts from A to J. 	(A) Synaptic knob (B) Axon terminal (C) Node of Ranvier (D) Myelin sheath (E) Axon (F) Schwann cell (G) Nucleus (H) Cell body (I) Nissl's granules (J) Dendrites

35.	Label the parts in the given diagram of axon terminal and synapse.  	(A) Axon (B) Axon terminal (C) Synaptic vesicles (D) Pre-synaptic membrane (E) Synaptic cleft (F) Post-synaptic membrane (G) Receptors
36.	Which part of human brain is most developed?	Cerebrum
37.	Which part of our central nervous system acts as a master clock?	Hypothalamus
38.	Part of brain which consists of fibre tracts that interconnect different regions of brain.	Pons
39.	The medulla of brain is connected to	Spinal cord
40.	The medulla contains centres which control_____, _____ and _____	Respiration, cardiovascular reflexes and gastric secretion
41.	Cerebellum has very convoluted surface. Why	In order to provide the additional space for many more neurons.

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Chemical Coordination and Integration

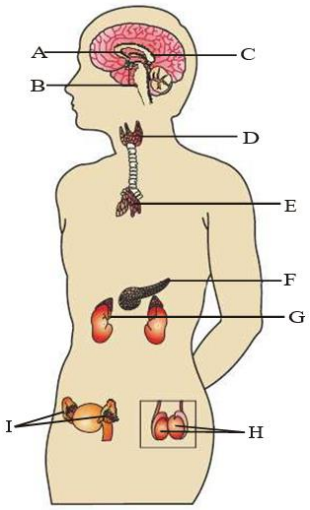
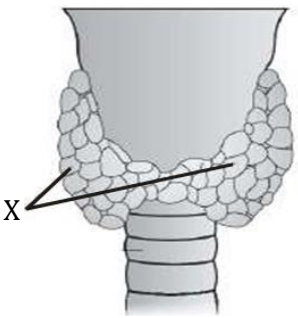
Q.NO.	QUESTIONS	ANSWERS
1.	Chemical which are non-nutrient and act as inter cellular messengers are known as.	Hormone
2.	Tissues located in different parts of body like pituitary, pineal, thyroid etc are included as.	Organised endocrine glands
3.	Group of neurosecretory cells which produces hormones and are located in hypothalamus are known as.	Nuclei
4.	Hormone which is responsible to inhibit the release of growth hormone from pituitary is.	Somatostatin
5.	Anterior pituitary & posterior pituitary hormones are regulated by hypothalamus respectively through.	Portal circulatory system & direct neural regulation
6.	Bony cavity in which pituitary gland is situated in	Sella tursica
7.	 'A' is the portion of adenohypophysis which secretes different hormones 'A' is _____.	Pars distalis
8.	Apart from humans melanocyte stimulating hormone is secreted through.	Pars intermedia
9.	Hormone which are released from posterior pituitory are :	Oxytocin and vasopressin
10.	Over secretion and low secretion of GH at childhood leads to respectively.	Gigantism and pituitary dwarfism
11.	Hormone secreted by pituitary gland and regulates the mammary gland and formation of milk in them is.	Prolactin
12.	In females, vigorous contraction of uterus at the time of child birth and milk ejection from the mammary gland is due to	Oxytocin
13.	Hormone responsible for reabsorption of water and electrolyte to reduce loss of water is	Antidiuretic hormone (ADH)
14.	Identify A, B, C and D, E. 	A–Anterior pituitary B–Posterior pituitary C–Hypothalamus D–Hypothalamic neuron E–Portal circulation

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15.	Gland which is responsible for rhythms of sleep wake cycle	Pineal gland
16.	Deficiency of I_2 in diet leads to	Goiter
17.	Stunted growth of child during pregnancy mental retardation, low intelligence quotient is due to	Hypothyrodism
18.	Hormone responsible for basal metabolic rate is	Thyroxine
19.	Hormone, responsible for raised level of Ca^{+2} in blood is	PTH
20.	Antagonist pair of hormone which regulate Ca^{+2} in blood are	PTH and TCT
21.	Hormone which play a major role in differentiation of T–Lymphocyte is	Thymosin
22.	Catecholamine which are secreted in emergency conditions are	Adrenaline and noradrenaline
23.	Main glucocorticoid hormone responsible for carbohydrate metabolism is	Cortisol
24.	Hormone which inhibit cellular uptake and utilisation of amino acid as well as does hyperglycemia through gluconeogenesis is	Cortisol
25.	Hormone of adrenal cortex which helps in maintaining electrolyte and body fluid volume is	Aldosterone
26.	Hormone which enhance cellular uptake and utilisation of glucose as well as does hypoglycemia through glycogenesis is	Insulin
27.	Male and female sex hormone respectively which stimulate development of sex organ, sexual behaviour, gametogenesis are	Testosterone and Estrogen
28.	Chemical nature of all the hormone which are secreted from pituitary and hypothalamus is	Proteineous
29.	Ca^{+2}, IP_3, C–AMP etc. are generated after binding of hormone on receptor present on membrane thus these are termed as	Secondary messenger
30.	Hormone which regulate gene expression or chromosome by interaction of hormone receptor complex with the genome are	Steroid and Iodothyronin hormone
31.	Endocrine glands lack ducts and are hence, called____ glands and their secretions are called _____.	Ductless, hormones
32.	_____ possess very simple endocrine systems with few hormones whereas a large number of chemicals act as hormones and provide coordination in the _____.	Invertebrates, vertebrates
33.	Hypothalamus is the basal part of _____.	diencephalon
34.	Name the hypothalamic hormone which stimulates the pituitary synthesis and release of gonadotrophins.	Gonadotrophic releasing hormone
35.	Label A, B, C, D, E, F, G, H and I is the given diagram of location of endocrine glands	(A) Hypothalamus (B) Pituitary (C) Pineal (D) Thyroid and parathyroid (E) Thymus

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		(F) Pancreas (G) Adrenal (H) Testis (I) Ovary
36.	Name the hormone which stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.	Thyroid stimulating hormone (TSH)
37.	LH and FSH are called gonadotrophins. Why?	Because they stimulates
38.	Name the hormone which stimulates the synthesis and secretion of androgens from testis.	LH (Luteinizing hormone)
39.	What is the function of LH in females?	LH in females induces ovulation of fully mature follicles and maintains the corpus luteum
40.	Name the hormone secreted by pars intermedia and its function.	Melanocyte stimulating hormone, regulates pigmentation of the skin
41.	 (A) Name the gland shown in the diagram (B) Name the hormones secreted by this gland (C) Which element is essential for the normal rate of hormone synthesis in this gland ?	(A) Thyroid (B)Thyroxine, Thyrocalcitonin (C) Iodine
42.	In adult women, _____ may cause menstrual cycle to become irregular.	Hypothyroidism
43.	I am a lobular structure located between lungs behind sternum on the ventral side of aorta and plays an important role in the development of the immune system. Identify me.	Thymus gland

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44.	Adrenaline and noradrenaline are called emergency hormones. Why?	Because these are secreted in response of any kind of stress and during emergency conditions.
45.	The adrenal cortex can be divided intro three layers, called _____ (inner layer), _____ (middle layer) and _____ (outer layer).	Zona reticularis, zona fasciculata, zona glomerulosa
46.	Name the main mineralocorticoid in our body.	Aldosterone
47.	Endocrine part of pancreas is called _____.	islets of Langerhans
48.	Alpha cells of islets of Langerhans secrete _____.	glucagon
49.	Beta cells of islets of Langerhans secrete _____.	insulin
50.	Write some symptoms of diabetes mellitus.	Glycosuria (loss of glucose through urine), ketonuria (loss of ketone bodies through urine)
51.	Name the hormone secreted by atria of heart which causes dilation of blood vessels and reduces blood pressure.	ANF (Atrial Natri Uretic Factor)
52.	The _____ of kidney produce a peptide hormone called_____ which stimulates _____.	Juxta glomerular cells, erythropoietin, erythropoiesis
53.	Name the hormone which increases the secretion of pepsinogen and hydrochloric acid from gastric glands.	Gastrin
54.	Which hormonal deficiency is responsible for the following ? (a) Diabetes mellitus (b) Goitre (c) Cretinism	(a) insulin (b) iodine (c) thyroxine
55.	Match the following columns : **Column A** / **Column B** a. T4 — i. Hypothalamus b. PTH — ii. Thyroid c. GnRH — iii. Pituitary d. LH — iv. Parathyroid	(a) ii (b) iv (c) i (d) iii
56.	Fill in the blanks. **Hormones** / **Target gland** a. Hypothalamic hormones _____ b. Thyrotrophin (TSH) _____ c. Corticotrophin (ACTH) _____ d. Gonadotrophins (LH. FSH) _____ e. Metanotrophin (MSH) _____	(a) pituitary gland (b) thyroid gland (c) adrenal cortex (d) gonads (e) melanocytes (melanin secreting cells)

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Animal Kingdom

Q.No.	Question	Answer
1.	Members of which phylum are commonly known as sponges?	Porifera
2.	Which type of symmetry is mostly found in sponges ?	Asymmetry
3.	Presence of water transport system is characteristic of which phylum ?	Porifera (Sponges)
4.	What is the name of pores through which water enters into the body of sponges ?	Ostia
5.	What is the name of pore through which water comes out from body of sponges ?	Osculum
6.	What is the name of central cavity found in the body of members of phylum porifera ?	Spongocoel (Paragastric cavity)
7.	What is the function of water canal system in sponges ?	Food gathering, respiratory exchange and removal of waste
8.	What is the name of cells those line the spongocoel ?	Choanocytes or collar cells
9.	Name the structures which support the body of sponges as a skeleton.	Spicules and spongin fibres
10.	The animals which don't have separate sexes are called as :	Hermaphrodite (Bisexual)
11.	When development of an organism occurs through larval stage, than this type of development is called as :	Indirect development
12.	Which type of symmetry is found in coelenterates ?	Radial symmetry
13.	Name the cell that is used for defence purposes in coelenterates.	Cnidoblast
14.	Which cnidarians possess a skeleton made up of calcium carbonate?	Corals
15.	The alternation of Polyp and Medusa forms in coelenterates is called as :	Metagenesis
16.	Which organism is also known as Portuguese man of war ?	*Physalia*
17.	Members of which phylum are commonly known as sea walnuts or comb jellies ?	Ctenophora
18.	Which structure helps in locomotion in ctenophores ?	Comb plates
19.	The property of a living organism to emit light is called as :	Bioluminescence
20.	Members of which phylum are known as flatworms ?	Platyhelminthes
21.	Which type of symmetry is found in the members of phylum platyhelminthes ?	Bilateral symmetry
22.	Structures which help in osmoregulation and excretion in flatworms are called_____.	Flame cells
23.	Members of which phylum are also known as roundworms ?	Aschelminthes (Nematoda)

24.	Name the phylum in which organ system level of body organisation appeared first time.	Aschelminthes (Nematoda)
25.	Which phylum of animals posses pseudocoelomic type of body cavity ?	Aschelminthes (Nematoda)
26.	Which structures help in osmoregulation and excretion in Annelids ?	Nephridia
27.	The body of annelids is distinctly marked out into segments. These segments are called as.	Metameres
28.	Name the largest phylum of animal kingdom.	Arthropoda
29.	The exoskeleton of arthropods is made up of _____.	Chitin
30.	The name of the balancing organ in arthropods.	Statocyst
31.	What is the main function of malpighian tubules ?	Excretion
32.	Name the organism which produces silk ?	*Bombyx* (Silk worm)
33.	Name organism which produces lac.	*Laccifer* (Lac insect)
34.	Name the second largest phylum of animal kingdom.	Mollusca
35.	The soft and spongy layer of skin over the visceral hump of molluscs is known as :	Mantle
36.	In which phylum the mantle cavity is found ?	Mollusca
37.	What is the name of file like rasping organ found in molluscs?	Radula
38.	Animals of which phylum are reared for the production of pearls ?	Mollusca
39.	In which phylum the spiny bodied organisms are included ?	Echinodermata
40.	In which phylum organisms alter their symmetry from bilateral to radial during the development ?	Echinodermata
41.	Members of which phylum have ventral mouth and dorsal anus ?	Echinodermata
42.	Water vascular system is a characteristic of which phylum ?	Echinodermata
43.	Name the excretory organ of *Balanoglossus.*	Proboscis gland
44.	Which subphylum are classified under protochordates ?	Urochordata and cephalochordata
45.	In which subphylum notochord is present only in larval tail ?	Urochordata
46.	In which subphylum notochord extends from head to tail and persistent throughout the life ?	Cephalochordata
47.	In which subphylum notochord is replaced by vertebral column ?	Vertebrata
48.	Animals of which class possess a sucking and circular mouth without jaws ?	Cyclostomata
49.	In which class of fishes mouth is located ventrally ?	Chondrichthyes
50.	Which types of scales are found on the body of cartilagenous fishes?	Placoid scales
51.	In which class of fishes notochord is persistent throughout the life ?	Chondrichthyes
52.	The teeth of cartilaginous fishes are formed due to modification of which type of scales ?	Placoid scales
53.	What is the name of structure which prevents the bony fishes from sinking ?	Air bladder
54.	Why cartilaginous fishes have to swim constantly ?	Because they lack air bladder

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55.	Name a fish which has electric organ.	*Torpedo*
56.	Name a fish which possess poisonous sting.	*Trygon*
57.	The animals which don't have the capacity to regulate their body temperature are called as _____.	Poikilothermous
58.	The animals which have the capacity to regulate their body temperature are called as _____.	Homeothermous
59.	How many pairs of gills are present in bony fishes ?	4-pairs
60.	Name the cap which covers the gills in osteichthyes fishes.	Operculum
61.	Which structure represents ear in amphibians and reptiles ?	Tympanum
62.	In which class of vertebrates 3-chambered heart is found ?	Amphibia and reptilia
63.	The common chamber for alimentary canal, urinary bladder and reproductive tracts is called as :	Cloaca
64.	Which class of vertebrates possesses dry and cornified skin with epidermal scales or scutes ?	Reptilia
65.	In which reptiles 4-chambered heart is present ?	Crocodiles
66.	Snakes and lizards shed off their scales. This phenomenon is called as :	Skin cast
67.	The presence of feathers is characteristic of which class ?	Aves
68.	The long bones of birds are hollow with air cavities. These types of bones are called as :	Pneumatic bones
69.	Name of the additional chambers found in the digestive tract of birds.	Crop and Gizzard
70.	Name the respiratory organs in birds and mammals.	Lungs
71.	Presence of mammary gland is the unique character of which class?	Mammalia
72.	Name an oviparous animal with presence of mammary glands.	*Ornithorhynchus* (Platypus)
73.	Name the phyla which have radial symmetry.	Coelenterata, Ctenophora, Echinodermata (adult)
74.	The undifferentiated layer present between ectoderm and endoderm in diploblastic animals is known as :	Mesoglea
75.	A body cavity in which mesoderm is found in scattered form is called as :	Pseudocoelom
76.	Match the columns : (A) *Sycon* (i) Bath sponge (B) *Spongilla* (ii) Scypha (C) *Euspongia* (iii) Fresh water sponge	 (A)-(ii) (B)-(iii) (C)-(i)
77.	Match the columns : (A) *Adamsia* (i) Sea fan (B) *Pennatula* (ii) Brain coral (C) *Gorgonia* (iii) Sea pen (D) *Meandrina* (iv) Sea anemone	 (A)-(iv) (B)-(iii) (C)-(i) (D)-(ii)

78.	Match the columns :		
	(A) *Taenia*	(i) Filaria worm	(A)-(iii)
	(B) *Fasciola*	(ii) Round worm	(B)-(v)
	(C) *Ascaris*	(iii) Tapeworm	(C)-(ii)
	(D) *Wuchereria*	(iv) Hook worm	(D)-(i)
	(E) *Ancylostoma*	(v) Liver fluke	(E)-(iv)
79.	Match the columns :		
	(A) *Pheretima*	(i) King crab	(A)-(iii)
	(B) *Hirudinaria*	(ii) Honey bee	(B)-(iv)
	(C) *Apis*	(iii) Earthworm	(C)-(ii)
	(D) *Limulus*	(iv) Blood sucking leech	(D)-(i)
80.	Match the columns :		
	(A) *Pila*	(i) Devil fish	(A)-(iii)
	(B) *Pinctada*	(ii) Chiton	(B)-(v)
	(C) *Sepia*	(iii) Apple snail	(C)-(vi)
	(D) *Loligo*	(iv) Tusk shell	(D)-(vii)
	(E) *Octopus*	(v) Pearl oyester	(E)-(i)
	(F) *Aplysia*	(vi) Cuttlefish	(F)-(viii)
	(G) *Dentalium*	(vii) Squid	(G)-(iv)
	(H) *Chaetopleura*	(viii) Sea hare	(H)-(ii)
81.	Match the columns :		
	(A) *Asterias*	(i) Brittle star	(A)-(iii)
	(B) *Echinus*	(ii) Sea lily	(B)-(iv)
	(C) *Antedon*	(iii) Star fish	(C)-(ii)
	(D) *Cucumaria*	(iv) Sea urchin	(D)-(v)
	(E) *Ophiura*	(v) Sea cucumber	(E)-(i)
82.	Match the columns :		
	(A) *Branchiostoma*	(i) Saw fish	(A)-(iii)
	(B) *Petromyzon*	(ii) Dog fish	(B)-(v)
	(C) *Myxine*	(iii) Lancelet	(C)-(vi)
	(D) *Scoliodon*	(iv) Sting ray	(D)-(ii)
	(E) *Pristis*	(v) Lamprey	(E)-(i)
	(F) *Carcharodon*	(vi) Hag fish	(F)-(vii)
	(G) *Trygon*	(vii) Great white shark	(G)-(iv)
83.	Match the columns :		
	(A) *Exocoetus*	(i) Angel fish	(A)-(v)
	(B) *Hippocampus*	(ii) Rohu	(B)-(vi)
	(C) *Labeo*	(iii) Fighting fish	(C)-(ii)
	(D) *Clarias*	(iv) Magur	(D) (iv)
	(E) *Betta*	(v) Flying fish	(E)-(iii)
	(F) *Pterophyllum*	(vi) Sea horse	(F)-(i)

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84.	Match the columns :		
	(A) *Bufo*	(i) Limbless amphibian	(A)-(iii)
	(B) *Rana*	(ii) Indian bull frog	(B)-(ii)
	(C) *Hyla*	(iii) Common toad	(C)-(iv)
	(D) *Ichthyophis*	(iv) Tree frog	(D)-(i)
85.	Match the columns :		
	(A) *Chelone*	(i) Krait	(A)-(iv)
	(B) *Testudo*	(ii) Wall lizard	(B)-(v)
	(C) *Chameleon*	(iii) Garden lizard	(C)-(vi)
	(D) *Calotes*	(iv) Turtle	(D)-(iii)
	(E) *Naja*	(v) Tortoise	(E)-(vii)
	(F) *Bangarus*	(vi) Tree lizard	(F)-(i)
	(G) *Hemidactylus*	(vii) Cobra	(G)-(ii)
86.	Match the columns :-		
	(A) *Corvus*	(i) Vulture	(A)-(ii)
	(B) *Columba*	(ii) Crow	(B)-(vi)
	(C) *Psittacula*	(iii) Penguin	(C)-(vii)
	(D) *Struthio*	(iv) Peacock	(D)-(v)
	(E) *Pavo*	(v) Ostrich	(E)-(iv)
	(F) *Aptenodytes*	(vi) Pigeon	(F)-(iii)
	(G) *Neophron*	(vii) Parrot	(G)-(i)
87.	Match the columns :		
	(A) *Ornithorhynchus*	(i) Flying fox	(A)-(v)
	(B) *Macropus*	(ii) Blue whale	(B)-(vii)
	(C) *Pteropus*	(iii) Monkey	(C)-(i)
	(D) *Balaenoptera*	(iv) Dog	(D)-(ii)
	(E) *Canis*	(v) Platypus	(E)-(iv)
	(F) *Macaca*	(vi) Cat	(F)-(iii)
	(G) *Felis*	(vii) Kangaroo	(G)-(vi)
	(H) *Equus*	(viii) Horse	(H)-(viii)
88.	Match the columns :		
	(A) Operculum	(i) Ctenophora	(A)-(viii)
	(B) Parapodia	(ii) Mollusca	(B)-(v)
	(C) Scales/Scutes	(iii) Porifera	(C)-(iv)
	(D) Comb plates	(iv) Reptilia	(D)-(i)
	(E) Radula	(v) Annelida	(E)-(ii)
	(F) Hairs	(vi) Cyclostomata	(F)-(vii)
	(G) Choanocytes	(vii) Mammalia	(G)-(iii)
	(H) Gill slits	(viii) Osteichthyes	(H)-(vi)
89.	Match the columns :		
	(A) Water canal system	(i) Coelenterata	(A)-(vii)
	(B) Metagenesis	(ii) Platyhelminthes	(B)-(i)
	(C) Comb plates	(iii) Mollusca	(C)-(viii)

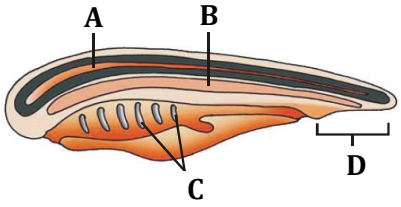
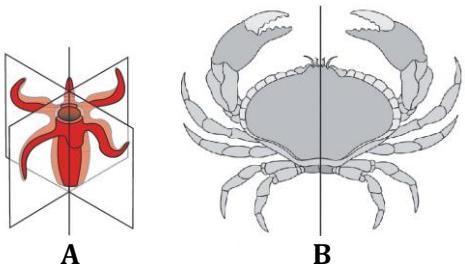
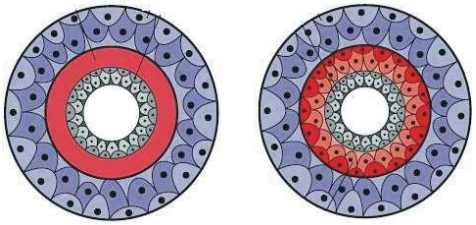
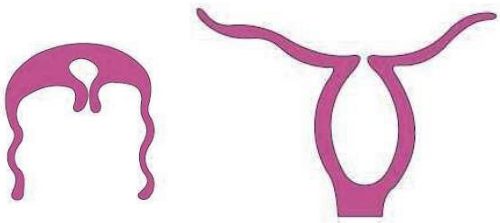
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	(D) Flame cells	(iv) Arthropoda	(D)-(ii)
	(E) Psedocoelom	(v) Echinodermata	(E)-(vi)
	(F) Jointed appendages	(vi) Aschelminthes	(F)-(iv)
	(G) Soft bodied animals	(vii) Porifera	(G)-(iii)
	(H) Water vascular system	(viii) Ctenophora	(H)-(v)
90.	Match the columns :		
	(A) Urochordata	(i) *Camelus*	(A)-(ii), (viii), (ix)
	(B) Hemichordata	(ii) *Ascidia*	(B)-(v), (xi)
	(C) Arthropoda	(iii) *Crocodilus*	(C)-(vii), (xiii), (xv)
	(D) Reptilia	(iv) *Elephas*	(D)-(iii), (xiv), (xvi)
	(E) Mammalia	(v) *Balanoglossus*	(E)-(i), (iv), (vi), (x), (xii)
		(vi) *Rattus*	
		(vii) *Culex*	
		(viii) *Salpa*	
		(ix) *Doliolum*	
		(x) *Delphinus*	
		(xi) *Saccoglossus*	
		(xii) *Panthera*	
		(xiii) *Aedes*	
		(xiv) *Vipera*	
		(xv) *Anopheles*	
		(xvi) *Alligator*	
91.	Mark the given statements True (T) or False (F) :		
	(i) All members of Animalia are multicellular.		(T)
	(ii) Sponges exhibit tissue grade level of body organization.		(F)
	(iii) A complete digestive system has two openings, mouth and anus.		(T)
	(iv) Annelids are pseudocoelomate animals.		(F)
	(v) Notochord is an endodermally derived structure.		(F)
	(vi) All vertebrates are chordates.		(T)
	(vii) All chordates are vertebrates.		(F)
	(viii) Polyps produce medusae asexually and medusae form the polyps sexually.		(T)
	(ix) Hooks and suckers are present in non parasitic flatworms.		(F)
	(x) Lamprey dies after few days of spawning.		(T)
92.	Fill in the blanks :		

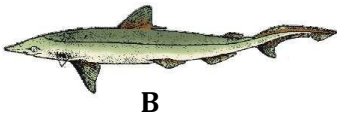
Character	Non Chordates	Chordates
Notochord	Absent	(a)
Central nervous system	Ventral	(b)
	(c)	Hollow
	Double	(d)
Gills slits	Absent	(e)
Heart	(f)	Ventral
Post anal tail	(g)	Present

(a) - Present
(b) - Dorsal
(c) - Solid
(d) - Single
(e) - Present
(f) - Dorsal
(g) - Absent

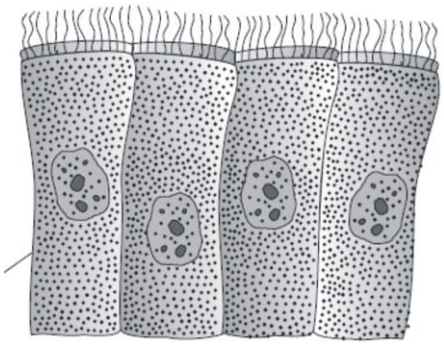
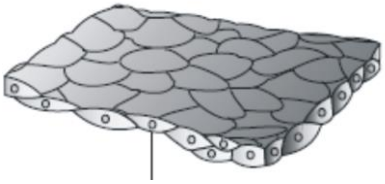
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93.	Identify the following A, B, C and D in given figure :  	(A) - Nerve cord (B) - Notochord (C) - Gill slits (D) - Post anal tail
94.	Identify the following symmetries shown in the figures A and B :  	(A)-Radial Symmetry (B)- Bilateral symmetry
95.	Identify the following diagrams A, B and C with respect to presence or absence of coelom.  **(A)** **(B)** **(C)**	(A) Coelomate (B) Pseudocoelomate (C) Acoelomate
96.	Identify the following body organizations on the basis of germinal layers :  **(A)** **(B)**	(A)-Diploblastic, (B)-Triploblastic
97.	Identify the following body forms shown in the figure :  **(A)** **(B)**	(A)- Medusa, (B)- Polyp

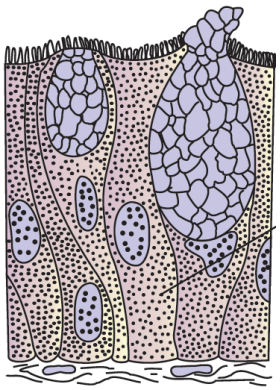
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98.	The given structure is found in phylum ___(a)___ and helps in _(b)___. 	(a)- Coelenterata, (b)- Anchorage, defense and capturing prey.
99.	Give the answer of following questions by observations from given figures :  **A**  **B**  **C**  **D** (i) Which fish has air bladder ? (ii) Which fish is known as saw fish ? (iii) Which fish is cartilaginous fish ? (iv) Which animal is ectoparasite on true fishes ? (v) Which fish has paired fins ? (vi) Which fish has bony skeleton ? (vii) Which fish has 4 - pairs of gills ? (viii) Which fish has operculum ? (ix) Which fishes are viviparous ? (x) Which fish lives in fresh water ?	(i) - D (ii) - C (iii) - B, C (iv) - A (v) - B, C, D (vi) - D (vii) - D (viii) - D (ix) - B, C (x) - D
100.	Select the correct options for given animals:  (i) Diploblastic / Triploblastic (ii) Acoelomate / Pseudocoelomate / Coelomate (iii) Chordate / Non Chordate (iv) Urochordate / Cephalochordate	(i) - Triploblastic (ii) - Coelomate (iii) - Chordate (iv) - Urochordate

Structural Organisation in Animals
(Animal Tissue, Frog, Cockroach)

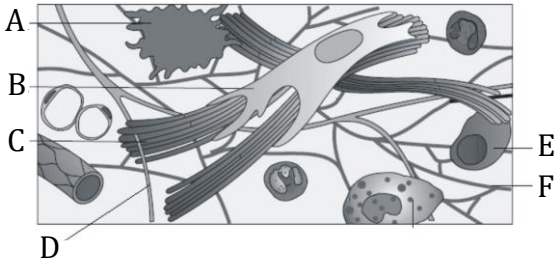
Q.No.	Question	Answer
1.	Mark the statement as true or false (a) In unicellular organisms all the functions are performed by single cell. (b) In the complex body of multicellular animals different functions are carried out by different group of cells. (c) In multicellular animals, a group of similar cells along with intercellular substances perform a specific function. Such an organisation is called tissue. (d) The functions of cells vary according to their structure .	 True True True False
2.	This tissue has a free surface, which faces either a body fluid or the outside environment and thus provides a covering or a lining for some part of the body. The cells are compactly packed with little intercellular matrix. This tissue is _____.	Epithelial tissue
3.	_____ epithelium is composed of a single layer of cells and functions as a lining for body cavities, ducts and tubes.	Simple
4.	_____ epithelium consists of two or more cell layers and has protective function as it does in our skin.	Compound
5.	 This epithelium is found in	Fallopian tubes , Brain ventricles
6.	 This epithelium is found in	Air sacs and blood vessels

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7.	The _____ epithelium is made of a single thin layer of flattened cells with irregular boundaries.	simple squamous
8.	The _____ epithelium is composed of a single layer of cube-like cells. This is commonly found in _____ and tubular parts of nephrons in kidneys and its main functions are _____ and _____.	Cuboidal, Ducts of glands, Secretion, Absorption
9.	The epithelium of _____ of nephron in the kidney has microvilli.	Proximal convoluted tubule
10.	Columnar epithelium is found in the walls of _____ and _____.	Stomach , intestine
11.	What is the function of Cilia ?	To move particles and mucus in a specific direction over epithelium
12.	Bronchioles and fallopian tubes are lined by _____.	Ciliated epithelium
13.	Some of the columnar or cuboidal cells get specialised for secretion and are called _____ epithelium.	glandular
14.	 Give an example of this gland	Goblet cells
15.	_____ glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products.	Exocrine
16.	_____ epithelium is made of more than one layer (multi-layered) of cells and thus has a limited role in _____.	Compound, Secretion and absorption
17.	Write the functions of compound epithelium.	To provide protection against chemical and mechanical stress
18.	_____ junctions help to stop substances from leaking across a tissue.	Tight
19.	_____ junctions perform cementing to keep neighbouring cells together.	Adhering

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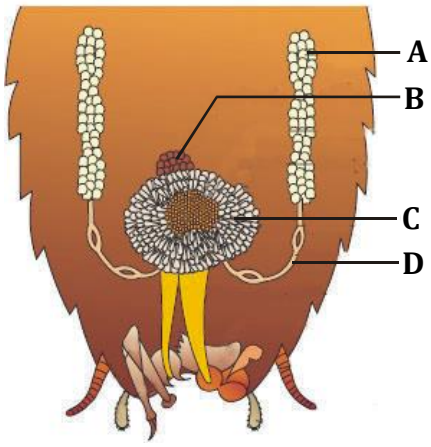
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20.	_____ junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.	Gap
21.	_____ tissues are most abundant and widely distributed in the body of complex animals.	Connective
22.	Connective tissues are named as connective tissues. Why?	Because of their special function of linking and supporting other tissues / organs of the body
23.	In all connective tissues except _____, the cells secrete fibres of structural proteins called collagen or elastin.	blood
24.	 Identify A, B, C, D, E and F in the given diagram: -	(A) MACROPHAGE (B) FIBROBLASTS (C) COLLAGEN FIBERS (D) ELASTIN FIBRES (E) LYMPHOCYTE (F) RETICULAR FIBRES
25.	Connective tissues are classified into three types : (i) _____ connective tissue, (ii) _____ connective tissue and (iii) _____ connective tissue.	(i) loose (ii) dense (iii) specialised
26.	_____ connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance, for example, areolar tissue present beneath the skin	Loose
27.	Loose connective tissue which store fat is	adipose
28.	In the _____ connective tissues, the collage fibres are present in rows between many parallel bundles of fibres _____ which attach skeletal muscles to bones and _____ which attach one bone to another are examples of this tissue.	Dense regular, tendons, ligaments
29.	The intercellular material of cartilage is _____ and _____ and resists compression.	solid, pliable
30.	Most of the cartilages in vertebrate embryos are replaced by _____ in adults.	bones
31.	Frogs can live both on land and in freshwater and belong to class Amphibia of phylum _____.	Chordata

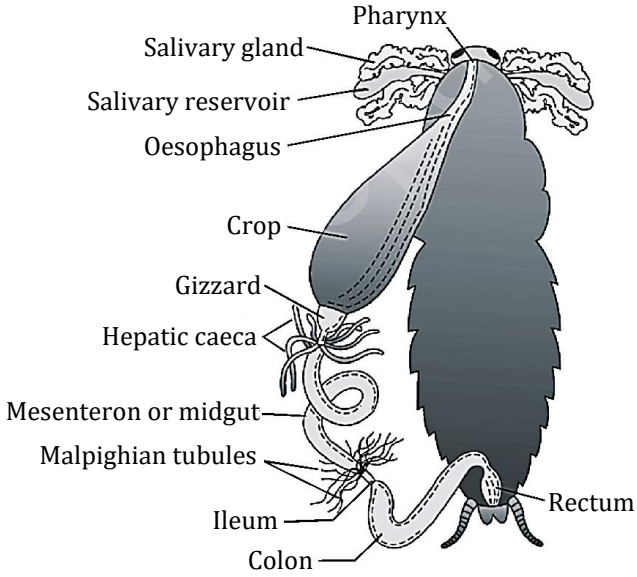
32.	The most common species of frog found in India is _____.	Rana tigrina
33.	Frogs do not have constant body temperature i.e., their body temperature varies with the temperature of the environment. Such animals are called	cold blooded or poikilotherms
34.	They have the ability to change the colour to hide them from their enemies (camouflage). This protective coloration is called _____	mimicry
35.	Frogs are not seen during peak summer and winter. During this period they take shelter in deep burrows to protect them from extreme heat and cold. This is known as summer sleep (_____) and winter sleep (_____) respectively	Aestivation, hibernation
36.	The skin of frog is smooth and slippery due to the presence of	mucus
37.	Body of a frog is divisible into	head and trunk
38.	Male frogs can be distinguished by the presence of sound producing _____and also a _____on the first digit of the fore limbs which are absent in female frogs.	vocal sacs, copulatory pad
39.	Food in frog is captured by the _____ tongue.	bilobed
40.	Frogs respire on land and in the water by two different methods. In water, skin acts as aquatic respiratory organ (_____respiration). Dissolved oxygen in the water is exchanged through the skin by diffusion. On land, the buccal cavity, skin and lungs act as the respiratory organs. The respiration by lungs is called _____ respiration.	Cutaneous, Pulmonary
41.	In frog during aestivation and hibernation gaseous exchange takes place through _____.	skin
42.	The vascular system of frog is well-developed _____ type	closed
43.	The frog excretes _____ and thus is a _____ animal	Urea , ureotelic
44.	In frog there are _____ pairs of cranial nerves arising from the brain.	ten
45.	Brain is enclosed in a bony structure called brain box (_____).	cranium
46.	In frog the brain is divided into	fore-brain, mid-brain and hind-brain
47.	In frog the midbrain is characterised by _____	A pair of optic lobes

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48.	Frog has different types of sense organs, namely organs of touch (_____), taste (_____), smell (_____), vision (_____) and hearing (_____)	Sensory papillae , taste buds , nasal epithelium, eyes , tympanum with internal ears
49.	Eyes in a frog are a pair of spherical structures situated in the _____ in skull	Orbit
50.	Frogs have well organised male and female reproductive systems. Male reproductive organs consist of a pair of yellowish ovoid testes, which are found adhered to the upper part of kidneys by a double fold of peritoneum called_____. Vasa efferentia are _____ in number that arise from testes. They enter the kidneys on their side and open into _____	Mesorchium , 10 - 12 , Bidders canal
51.	The female reproductive organs include a pair of _____. The ovaries are situated near kidneys and there is no functional connection with kidneys. A pair of oviduct arising from the ovaries opens into the cloaca separately. A mature female can lay _____ at a time	Ovaries , 2500 - 3000 ova
52.	Fertilisation in frog is _____ and takes place in water	external
53.	Study the given figure of male reproductive system of cockroach. In which of the following part (A, B, C and D) sperms are formed ?  	(A)
54.	How many malpighian tubules present in cockroach ?	100-150
55.	**Fill in the blank:-** In female cockroach sperms are transferred through _____ by male cockroach?	Spermatophore

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56.	Given below the diagram of alimentary canal of cockroach which type of structure responsible for storing of food.  	Crop
57.	In cockroach the respiratory system consists of a network of trachea, that open outside the body by the help of	Spiracles
58.	The number of thoracic segments in cockroach are :-	3
59.	The two pairs of wings in cockroach are situated in the segments.	Mesothorax and metathorax
60.	Which one of the following regions of the alimentary canal in Periplaneta is characterised by the presence of a peritrophic membrane?	Mid-gut
61.	In cockroach the nerve cord is :-	Double, ventral and solid
62.	In the mouth parts of cockroach the organ of mastication is :-	Mandibles
63.	A ring of 6-8 blind tubules called hepatic or gastric caecae is present at the junction of foregut and mid-gut in cockroach. It secrete the :-	Digestive juices
64.	Which of the following statements are true, which are false for cockroach (1) Sexual dimorphism is distinct (2) The anal styles are present on 9th sternum of male only (3) In both sexes the 10th segment bears anal cerci (4) Brain absent in cockroach	(1) $\rightarrow$ T, (2) $\rightarrow$ T, (3) $\rightarrow$ T, (4) $\rightarrow$ F
65.	Mushroom gland a part of :- (1) Digestive system of cockroach (2) Respiratory system of cockroach (3) Male reproductive system of earthworm (4) Male reproductive system of cockroach	(4)

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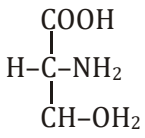
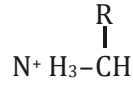
66.	Which type of structure in cockroach monitoring the environmental conditions :-	Antennae
67.	In male and female cockroach 10th segment bears a pair of jointed filamentous structure called :-	Anal cerci
68.	Body cavity of cockroach is called :-	Haemocoel
69.	Cockroach mainly excretes :-	Uric acid
70.	In female cockroach, ootheca is secreted by :-	Colleterial gland
71.	Egg of cockroach is :-	Centrolecithal and Megalecithal
72.	On the basis of metamorphosis the cockroach is :-	Paurometabolus
73.	A dorsal sclerite of cockroach is called :-	Tergum or Tergites
74.	In cockroach alary muscles are associated with :-	Heart and blood circulation
75.	Young one of the cockroach is called :-	Nymph
76.	The functional units of compound eye of cockroach are called :-	Ommatidia
77.	Ten pairs of spiracles for respiration are present in :-	Cockroach
78.	The scientific name of common cockroach is _____ .	*Periplaneta americana*
79.	In each segment of cockroach, exoskeleton has hardened plates called _____ .	Sclerites
80.	sclerites (tergites dorsally and sternites ventrally) that are joined to each other by a thin and flexible _____.	Articular membrane
81.	The mouth part of cockroach is _____ type.	Biting and chewing
82.	The first pair of wings arises from _____ and second pair from _____ in cockroach.	Mesothorax, Metathorax
83.	Mesothoracic wing in cockroach is also called as _____ .	Tegmina
84.	Male cockroach bear a pair of short, thread like _____, which are absent in female.	Anal styles
85.	In both sexes in cockroach, the 10th segment bears a pair of jointed filamentous structures called_____ .	Anal cerci
86.	In cockroach, _____ ganglia lie in the thorax, and _____ in the abdomen.	Three, Six
87.	Each eye in cockroach consists of about 2000 hexagonal _____ and gives _____vision.	Ommatidia, Mosaic

88.	The chitinous asymmetrical structure surrounding the gonopore is called _____ in cockroach.	Gonapophysis or phallomere
89.	The nymph grows by moulting about _____.	13 times
90.	In female cockroach, each ovary is formed of a group of _____ ovarioles.	8
91.	The fertilized eggs are encased in capsules called _____ in cockroach.	Ootheca
92.	On an average, female cockroach produce _____ oothecae, each containing _____eggs.	9-10 ; 14-16
93.	Summer sleep is called _____ whereas winter sleep is called _____ in animals such as frog.	Aestivation, Hibernation
94.	_____ protects eyes of frog in water.	Nictitating membrane
95.	Frog has _____ chambered heart containing _____ atria and _____ ventricle.	Three, Two, One
96.	The ventricle of heart in frog opens into a sac-like conus arteriosus on _____ side of heart.	Ventral side
97.	Frog excretes _____ as nitrogenous waste, thus is a _____ animal.	Urea, Ureotelic
98.	In frog there are _____ pair of cranial nerves arising from the brain.	Ten
99.	Male reproductive organs in frog are found adhered to upper part of kidney by a double fold of peritoneum called _____. Vasa efferentia enter the kidney and opens into _____.	Mesorchium, Bidder's canal
100.	In frog _____ is a small, median chamber that is used to pass faecal matter, urine and sperm to the exterior.	Cloaca

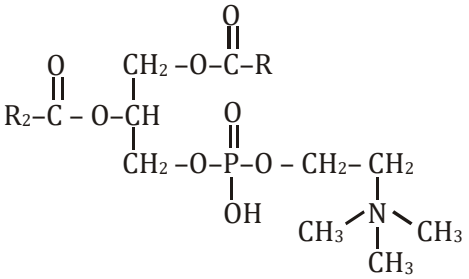
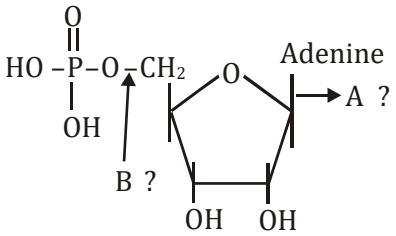
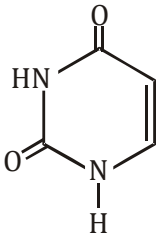
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Biomolecules

<table>
<tr><th>Q.No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1.</td><td>For the extraction of biomolecule from living tissue. This tissue has to be grind in which chemical?</td><td>Trichloroacetic acid (Cl_3CCOOH)</td></tr>
<tr><td>2.</td><td>If the tissue is fully burnt the remaining ash contain _____.</td><td>Inorganic elements</td></tr>
<tr><td>3.</td><td>In amino acid, the central carbon which contains an amino group and acidic group is called _____.</td><td>α-carbon</td></tr>
<tr><td>4.</td><td>Amino acids are substituted _____.</td><td>Methanes</td></tr>
<tr><td>5.</td><td>A variable group is attached on α-carbon in amino acid called ___.</td><td>–R group</td></tr>
<tr><td>6.</td><td>How many amino acids are used in protein synthesis?</td><td>20</td></tr>
<tr><td>7.</td><td>If the –R group in proteinaceous amino acid is hydrogen then the amino acid is _____.</td><td>Glycine</td></tr>
<tr><td>8.</td><td>Find the name of following amino acid
  </td><td>Serine</td></tr>
<tr><td>9.</td><td>Match the columns
<table>
<tr><th></th><th>A</th><th></th><th>B</th></tr>
<tr><td>(i)</td><td>Glutamic acid</td><td>(a)</td><td>Basic amino acid</td></tr>
<tr><td>(ii)</td><td>Valine</td><td>(b)</td><td>Acidic amino acid</td></tr>
<tr><td>(iii)</td><td>Lysine</td><td>(c)</td><td>Neutral amino acid</td></tr>
</table></td><td>(i)-b, (ii)-c, (iii)-a</td></tr>
<tr><td>10.</td><td>Match the columns
<table>
<tr><th></th><th>(A) Element</th><th></th><th>(B) % of weight in human body</th></tr>
<tr><td>(i)</td><td>Carbon (C)</td><td>(a)</td><td>65.0</td></tr>
<tr><td>(ii)</td><td>Hydrogen (H)</td><td>(b)</td><td>18.5</td></tr>
<tr><td>(iii)</td><td>Oxygen (O)</td><td>(c)</td><td>0.5</td></tr>
<tr><td>(iv)</td><td>Sodium (Na)</td><td>(d)</td><td>0.2</td></tr>
</table></td><td>(i)-b, (ii)-c, (iii)-a, (iv)-d</td></tr>
<tr><td>11.</td><td>  – given structure represent which form of amino acid?</td><td>Zwitter ionic form</td></tr>
<tr><td>12.</td><td>Lipids are generally ____ in water (soluble / insoluble)</td><td>Insoluble</td></tr>
<tr><td>13.</td><td>A fatty acid has two groups, –R group and ____.</td><td>Carboxyl group (–COOH)</td></tr>
<tr><td>14.</td><td>How many number of carbon atoms are present in palmitic acid including carboxyl carbon ?</td><td>16</td></tr>
<tr><td>15.</td><td>If the fatty acid contains one or more double bonds between C–C atoms then it is _____.</td><td>Unsaturated fatty acid</td></tr>
</table>

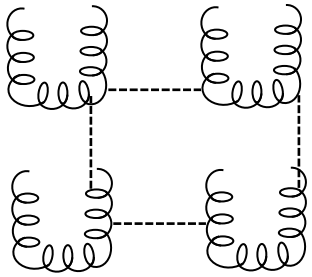
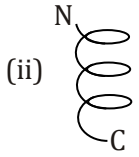
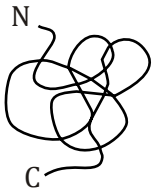
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<table>
<tr><td>16.</td><td>Trihydroxy propane is also called _____.</td><td>Glycerol</td></tr>
<tr><td>17.</td><td>Lipids are classified into fat or oil on the basis of their _____.</td><td>Melting point</td></tr>
<tr><td>18.</td><td>In monoglycerides, glycerol attached with fatty acid molecule by which bond ?</td><td>Ester bond</td></tr>
<tr><td>19.</td><td>The melting point of oil is _____ (low / high)</td><td>Low</td></tr>
<tr><td>20.</td><td>Which type of lipid is found in cell membrane?</td><td>Phospholipid</td></tr>
<tr><td>21.</td><td>Write the name of following lipid
  </td><td>Lecithin (Phospholipid)</td></tr>
<tr><td>22.</td><td>Living organisms have a number of carbon compounds in which heterocyclic rings can be found. Some of these are nitrogenous bases _____.</td><td>Adenine, thymine, guanine, cytosine, uracil</td></tr>
<tr><td>23.</td><td>Nitrogenous base + sugar = _____.</td><td>Nucleoside</td></tr>
<tr><td>24.</td><td>Nitrogenous base + sugar + phosphate = _____.</td><td>Nucleotide</td></tr>
<tr><td>25.</td><td>Identify the following bonds –
  </td><td>A = N-Glycosidic bond,
B = Ester bond</td></tr>
<tr><td>26.</td><td>Match the columns
<table>
<tr><th></th><th>(A)</th><th></th><th>(B)</th></tr>
<tr><td>(i)</td><td>Thymidine</td><td>(a)</td><td>Simplest amino acid</td></tr>
<tr><td>(ii)</td><td>Guanylic acid</td><td>(b)</td><td>Lipid</td></tr>
<tr><td>(iii)</td><td>Cholesterol</td><td>(c)</td><td>Nucleotide</td></tr>
<tr><td>(iv)</td><td>Glycine</td><td>(d)</td><td>Nucleoside</td></tr>
</table></td><td>(a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)</td></tr>
<tr><td>27.</td><td>Identify the following compound
  </td><td>Uracil N-base (pyrimidine)</td></tr>
</table>

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<table>
<tr><td>28.</td><td>Match the columns<table><tr><th></th><th>(A)</th><th></th><th>(B)</th></tr><tr><td>(i)</td><td>Pigment</td><td>(a)</td><td>Morphine</td></tr><tr><td>(ii)</td><td>Toxin</td><td>(b)</td><td>Ricin</td></tr><tr><td>(iii)</td><td>Drug</td><td>(c)</td><td>Carotenoids</td></tr><tr><td>(iv)</td><td>Alkaloids</td><td>(d)</td><td>Vinblastin</td></tr></table></td><td>(a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)</td></tr>
<tr><td>29.</td><td>Molecular weight of biomicromolecule approx. _____. (Less than 1000 dalton/more than 1000 dalton)</td><td>Less than 1000 dalton</td></tr>
<tr><td>30.</td><td>Match the columns<table><tr><th></th><th>(A) Protein</th><th></th><th>(B) Function</th></tr><tr><td>(a)</td><td>Collagen</td><td>(i)</td><td>Hormone</td></tr><tr><td>(b)</td><td>GLUT-4</td><td>(ii)</td><td>Intercellular ground substance</td></tr><tr><td>(c)</td><td>Insulin</td><td>(iii)</td><td>Glucose transport into cell</td></tr></table></td><td>(a)-(ii), (b)-(iii), (c)-(i)</td></tr>
<tr><td>31.</td><td>In a polypeptide chain amino acids are linked together by which bond ?</td><td>Peptide bond</td></tr>
<tr><td>32.</td><td>Most abundant protein in animal world is _____.</td><td>Collagen</td></tr>
<tr><td>33.</td><td>Polysaccharide that acts as store house of energy in plant tissue is _____.</td><td>Starch</td></tr>
<tr><td>34.</td><td>Inulin is a polymer of _____.</td><td>Fructose</td></tr>
<tr><td>35.</td><td>True/ False
(i) A protein is a homopolymer not a heteropolymer
(ii) RUBISCO is the most abundant protein in the whole biosphere
(iii) Cellulose is a homopolymer
(iv) In polysaccharide chain (glycogen) the right end is called non-reducing end
(v) Starch has helical structure and it given iodine test
(vi) Cellulose contains a complex helical structure and can hold I_2
(vii) Chitin is a heteropolymer</td><td>
False
True
True
False

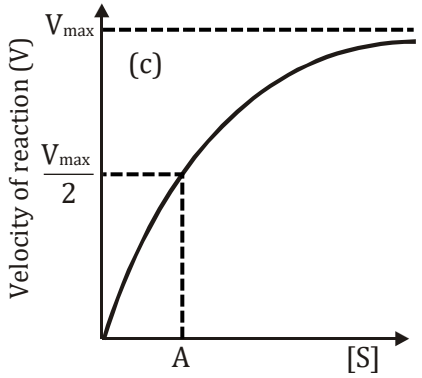
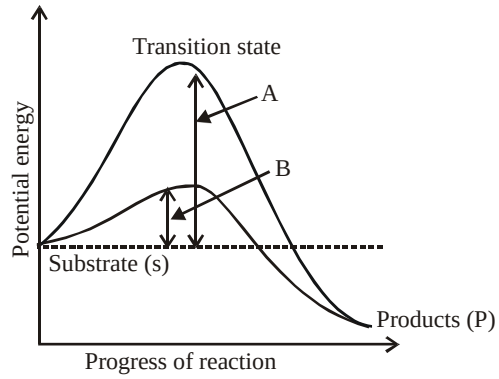
True
False
False</td></tr>
<tr><td>36.</td><td>Exoskeleton of Arthropods is made up of _____.</td><td>Chitin</td></tr>
<tr><td>37.</td><td>deoxyribose sugar is present in _____.</td><td>DNA</td></tr>
<tr><td>38.</td><td>Peptide bonds are formed between which groups of amino acids?</td><td>–COOH & $–NH_2$</td></tr>
<tr><td>39.</td><td>The glycosidic bonds of polysaccharide are formed by which process?</td><td>Dehydration synthesis</td></tr>
<tr><td>40.</td><td>The backbone of DNA chain is formed by _____.</td><td>Sugar-phosphate-sugar chain</td></tr>
<tr><td>41.</td><td>How many steps are involved in one complete turn of B-DNA?</td><td>10</td></tr>
<tr><td>42.</td><td>The distance per base pair in B-DNA is _____.</td><td>3.4 Å</td></tr>
<tr><td>43.</td><td>Which structure provides the positional information of amino acid in a protein ?</td><td>Primary structure</td></tr>
</table>

44.	The first amino acid in peptide chain is also called _____.	N-Terminal amino acid
45.	Which protein form is necessary for the many biological activities?	Tertiary structure
46.	When more than one polypeptide chains of tertiary structure are assembled together then called _____.	Quaternary structure
47.	In adult human haemoglobin, how many protein chains are present?	4 chains (2α & 2β)
48.	Guanine is always base pair with _____.	Cytosine
49.	Give one example of sulphur containing amino acid	Cysteine
50.	Match the following protein structure with their name (A) Structures (i)  (ii)  (iii) N——— C (iv)  (B) Name (a) Primary structure (b) Secondary structure (c) Tertiary structure (d) Quaternary structure	**A B** i d ii b iii a iv c
	Fill in the blanks	**Answer**
1.	Some nucleic acids that behave like enzymes, are called _____.	Ribozymes
2.	An active site of an enzyme is a ___ into which of the substrate fits.	crevice or pocket
3.	Inorganic catalysts work efficiently at ____, while enzymes get damaged at _____.	High temperature, high temperature
4.	_____ is thus an important quality of those enzymes which isolated from thermophilic organisms.	Thermal stability
5.	The substrate has to diffuse towards the 'active site.' There is thus, an _____ formation of a 'ES' complex.	Obligatory
6.	Inhibition of succinic dehydrogenase by malonate which closely resembles the substrate _____ in structure.	Succinate
7.	All enzymes catalysing inter-conversion of optical, geometric or positional isomers, are known as ____.	Isomerases
8.	Prosthetic groups are _____ compounds and are distinguished from other cofactor in that they are _____ bound to the apoenzyme.	organic, tightly
9.	High temperature destroys enzymatic activity because proteins are _____ by heat.	Denatured

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10.	When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as _____.	competitive inhibitor
11.	Enzymes catalysing the linking together of two compounds are known as _____.	Ligases
12.	Haem is the prosthetic group in _____ and _____, which catalyse the breakdown of hydrogen peroxide to water and oxygen.	Peroxidase, catalase
13.	Enzymes are divided into _____ classes each with _____ subs classes and named accordingly by a _____ digit number.	6, 4-13, four
14.	Each enzymes shows the highest activity at a particular temperature and pH called the ___ temperature and ____ pH respectively.	optimum, optimum
15.	During the state where substrate is bound to the enzyme active site, a new structure of the substrate called _____ structure is formed.	Transition state
16.	Enzymes eventually bring ___ the activation energy barrier making the transition of 'S' to 'P' more easy.	Down
	True/False	**Answer**
1.	Rate can also be called velocity if the direction is specified.	True
2.	Co-factor play a crucial role in the catalytic activity of the enzyme.	True
3.	Transferases catalysing a transfer of hydrogen between a pair of substrate S and S'.	False
4.	Co-enzymes are organic compounds but their association with the apoenzyme is only transient.	True
5.	Zinc is a cofactor for the proteolytic enzyme carboxypeptidase.	True
6.	The power of enzymes is incredible indeed.	True
7.	In our skeletal muscle under aerobic conditions, lactic acid is formed.	False
8.	Almost all enzymes are proteins.	True
9.	Stability is something related to energy status of the molecules or the structure.	True
10.	Coenzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin.	True
	Questions	**Answer**
1.	Which enzymes catalyse hydrolysis of ester, ether, peptide and glycosidic bonds?	Hydrolases
2.	Which enzymes catalyse, removal of groups from substrates by mechanisms other then hydrolysis leaving double bonds?	Lyases

	Diagram Based Questions	Answer
1.	Here A represent? 	Km
2.		A - Activation energy without enzyme. B - Activation energy with enzyme.

	Questions	Answer
1.	How many carbon atoms are present in stearic acid?	18
2.	Cellulose is a polymer of	β-glucose
3.	What happens on increasing the concentration of substrate in competitive inhibition?	Product is formed. (Inhibition is reversible)
4.	What happens to the Km value in competitive inhibition?	Km value increases.
5.	Which vitamin forms NAD and NADP?	Niacin
6.	Which metal ion is required as a cofactor for proteolytic enzyme carboxypeptidase?	Zn
7.	Which structure is absolutely necessary for the many biological activities of proteins?	3° STRUCTURE
8.	Each enzyme shows its highest activity at particular temperature and pH called the _____ temperature and _____ pH.	Optimum, optimum
9.	Zn is an cofactor of _____ enzyme.	Carboxypeptidase
10.	In glycogen, which kind of linkage is found between adjacent glucose molecule?	1,4 glycosidic linkage & 1,6 glycosidic linkage

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Human Reproduction

Q.NO.	QUESTION	ANSWER
1.	Male reproductive system is located in	Pelvic region
2.	Scrotum helps in maintaining low temperature of the testes about _____ lesser than body temperature.	2–2.5°C
3.	Each testis is _____ in shape.	Oval
4.	Length of each testis is __A__ in the length and __B__in width.	A = 4 to 5 cm, B = 2 to 3 cm
5.	Each testis has how many lobules ?	250
6.	The _____ undergo meiotic divisions finally leading to sperm formation.	Germ cells
7.	Androgens are secreted by	Leydig cells
8.	Which is the correct pathway of sperms. (i) Epididymis (ii) Seminiferous tubules (iii) Vas-efferentia (iv) Rete-testis	ii – iv – iii – i
9.	Glans penis is covered by a loose fold of skin is called.	Fore skin
10.	The male accessory glands include paired __A__, a __B__and paired __C__.	A-Seminal vesicle, B- prostate C- Bulbourethral glands
11.	Secretions of _____ glands also helps in the lubrication of penis.	Bulbourethral
12.	Primary spermatocyte is haploid.	False (Diploid)
13.	Secondary spermatocytes undergo first meiotic division	False (It show second meiotic division)
14.	Spermatids are transformed into spermatozoa by the process called.	Spermiogenesis
15.	Releasing of sperms from seminiferous tubules is known as.	Spermiation
16.	Spermatogenesis process start before birth.	False (At puberty due to availability of more GnRH)
17.	LH acts at the __A__cells and stimulates synthesis of __B__.	A- Leydig, B-Androgens
18.	FSH acts on the __A__cells and stimulates secretion of __B__which help in the process of __C__.	A-Sertoli, B-Some factors, C-Spermiogenesis
19.	The sperm head contains an elongated_____ nucleus.	Haploid
20.	Acrosomal enzyme helps in.	Fertilization of ovum
21.	Secretions of which parts are essential for maturation and motility of sperms ?	Epididymis, Seminal vesicle, prostate
22.	Sperms + ? = Semen.	Seminal plasma
23.	Functions of male sex accessory ducts and glands are maintained by	Androgens

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<table>
<tr><td>24.</td><td>Which of the following is not a part of female reproductive system?
Ovary, oviduct, uterus, vas deferens, hymen</td><td>Vas deferens</td></tr>
<tr><td>25.</td><td>Ovaries are the _____ female sex organs.</td><td>Primary</td></tr>
<tr><td>26.</td><td>Ovaries produce the (A) and several (B) hormones.</td><td>(A) female gamete (B) steroid</td></tr>
<tr><td>27.</td><td>Each ovary is about 5 to 8 cm in length in human. (True/false)</td><td>False</td></tr>
<tr><td>28.</td><td>The ovarian stroma is divided into two zones – a peripheral (A) and an inner (B) .</td><td>(A) Cortex (B) Medulla</td></tr>
<tr><td>29.</td><td>The oviduct, uterus and vagina constitute the female external genitalia. (True/false)</td><td>False</td></tr>
<tr><td>30.</td><td>Each fallopian tube is about _____ long.</td><td>10-12 cm</td></tr>
<tr><td>31.</td><td>Which help in collection of ovum after ovulation ?</td><td>Fimbriae</td></tr>
<tr><td>32.</td><td>The shape of the uterus is like an apple. (True/false)</td><td>False</td></tr>
<tr><td>33.</td><td>How many layers are present in uterus ?
One/Two/Three/Four</td><td>Three</td></tr>
<tr><td>34.</td><td>Correct sequence of uterine wall outer to inner side ?</td><td>Perimetrium → Myometrium → Endometrium</td></tr>
<tr><td>35.</td><td>The (A) undergoes cyclic changes during menstrual cycle while the (B) exhibits strong contraction during delivery of the baby.</td><td>(A) Endometrium
(B) Myometrium</td></tr>
<tr><td>36.</td><td>Which of the following does not include in external genitalia in female ?
Mons pubis, labia majora, fimbriae, hymen</td><td>Fimbriae</td></tr>
<tr><td>37.</td><td>Match the column :
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>(a)</td><td>Mons pubis</td><td>(i)</td><td>paired folds of tissue</td></tr>
<tr><td>(b)</td><td>Labia majora</td><td>(ii)</td><td>tiny finger like structure</td></tr>
<tr><td>(c)</td><td>Clitoris</td><td>(iii)</td><td>cushion of fatty tissue</td></tr>
<tr><td>(d)</td><td>Labia minora</td><td>(iv)</td><td>fleshy folds of tissue</td></tr>
</table></td><td>a-(iii), b-(iv), c-(ii), d-(i)</td></tr>
<tr><td>38.</td><td>The clitoris is a tiny finger like structure which lies at the upper junction of the (A) above the (B) .</td><td>(A) two labia minora
(B) urethral opening</td></tr>
<tr><td>39.</td><td>The presence or absence of hymen is not a reliable indicator of virginity or sexual experience. (True/false)</td><td>True</td></tr>
<tr><td>40.</td><td>A functional mammary gland is characteristic of all animal female. (True/false)</td><td>False</td></tr>
<tr><td>41.</td><td>The glandular tissue of each breast is divided into _____ mammary lobes.</td><td>15-20</td></tr>
<tr><td>42.</td><td>The cells of _____ secrete milk.</td><td>Alveoli</td></tr>
<tr><td>43.</td><td>The alveoli open into _____.</td><td>Mammary tubules</td></tr>
</table>

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<table>
<tr><td>44.</td><td>Several mammary ducts join to form a wider (A) which is</td><td>(A) mammary ampulla</td></tr>
<tr><td></td><td>connected to (B) through which milk is sucked out.</td><td>(B) lactiferous duct</td></tr>
<tr><td>45.</td><td>The primary sex organs the (A) in the males and the (B) in the females.</td><td>(A) testis
(B) ovaries</td></tr>
<tr><td>46.</td><td>Correct sequence of spermatogenesis :
(a) spermatogonia
(b) spermatids
(c) primary spermatocyte
(d) secondary spermatocyte
(e) spermatozoa</td><td>a → c → d → b → e</td></tr>
<tr><td>47.</td><td>After spermiogenesis; sperm head become embedded in the _____.</td><td>Sertoli cells</td></tr>
<tr><td>48.</td><td>Match the column :
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>(a)</td><td>Spermiation</td><td>(i)</td><td>Released of sperms from the seminiferous tubules</td></tr>
<tr><td>(b)</td><td>Insemination</td><td>(ii)</td><td>Semen is released by the penis into the vagina</td></tr>
<tr><td>(c)</td><td>Ejaculation</td><td>(iii)</td><td>Released of sperms outside the male body</td></tr>
</table></td><td>a-(i), b-(ii), c-(iii)</td></tr>
<tr><td>49.</td><td>Match the column :-
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>(a)</td><td>Spermatogonia</td><td>(i)</td><td>multiply by mitotic division</td></tr>
<tr><td>(b)</td><td>Primary spermatocyte</td><td>(ii)</td><td>undergo second meiotic division</td></tr>
<tr><td>(c)</td><td>Secondary spermatocytes</td><td>(iii)</td><td>undergo meiosis-I</td></tr>
<tr><td>(d)</td><td>Spermatids</td><td>(iv)</td><td>Transformed into sperms by the process of spermiogenesis</td></tr>
</table></td><td>a-(i), b-(iii), c-(ii), d-(iv)</td></tr>
<tr><td>50.</td><td>How many structures are haploid ?
Spermatogonia, Spermatozoa, Spermatids, Primary spermatocyte, Secondary spermatocytes</td><td>3 (Spermatozoa, Spermatid, Secondary spermatocyte)</td></tr>
<tr><td>51.</td><td>How many structures are diploid ?
Sperms, spermatid, spermatogonia, primary spermatocyte, secondary spermatocytes</td><td>2 (Spermatogonia, primary spermatocyte)</td></tr>
<tr><td>52.</td><td>Spermatogenesis starts at the age of (A) due to significant increase in the secretion of (B).</td><td>(A) Puberty (B) GnRH</td></tr>
<tr><td>53.</td><td>The increased levels of A acts at the B and stimulates secretion of two gonadotropis C.</td><td>(A) GnRH
(B) Anterior pituitary gland
(C) LH and FSH</td></tr>
</table>

54.	The sperm head contains an elongated diploid nucleus. (True/false)	False
55.	Which part of sperm produce energy for the movement of tail?	Middle piece
56.	The human male ejaculates about (A) million sperms during a coitus of which, for normal fertility, at least (B) percent sperms must have normal shape and size and at least (C) percent of them must show vigorous motility.	(A) 200-300 (B) 60 (C) 40
57.	The functions of male sex accessory ducts and glands are maintained by the _____.	Testicular hormones (Androgens)
58.	The process of formation of a mature female gamete is called ?	Oogenesis
59.	The oogenesis is initiated during the ..... stage.	Embryonic development
60.	How many primary follicles are left in each ovary at puberty ?	60,000 – 80,000
61.	Correct sequence of follicular development : (a) Graafian follicle (Mature tertiary) (b) Primary follicle (c) Tertiary follicle (d) Secondary follicle	b → d → c → a
62.	The Graafian follicle ruptures to release the (A) from the ovary by the process called (B).	(A) Secondary oocyte (B) Ovulation
63.	The reproductive cycle in the female primates is known as	Menstrual cycle
64.	The first menstruation begins at (A) and is called (B).	(A) Puberty (B) Menarche
65.	In human females, menstruation is repeated at an average interval of about ?	28 or 29 days
66.	How many ovum is released during the middle of each menstrual cycle ?	One
67.	Menstrual phase is completed about ?	3-5 days
68.	The menstrual flow results due to breakdown of	Endometrial lining
69.	The menstrual phase is followed by the which phase ?	Follicular phase
70.	During follicular phase changes in the ovary and the uterus are induced by changes in the levels of (A) and (B) hormones.	(A) Pituitary (B) Ovarian
71.	Both LH and FSH attain a peak level about ..... of menstrual cycle.	mid - cycle
72.	Which hormone is mainly responsible for ovulation ?	LH
73.	Which hormone is essential for maintenance of endometrium?	Progesterone
74.	In human beings, menstrual cycles ceases around the age of	50 years
75.	The process of fusion of a sperm with an ovum is called ?	Fertilisation

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<table>
<tr><td>76.</td><td>After completion of second meiotic division of the secondary oocyte and results in the formation of a (A) and a haploid (B).</td><td>(A) Second polar body
(B) Ovum</td></tr>
<tr><td>77.</td><td>How many chromosomes will be there in the zygote of human?</td><td>46</td></tr>
<tr><td>78.</td><td>It is correct to say that the sex of the baby is determined by the father and not by the mother. (False/true)</td><td>True</td></tr>
<tr><td>79.</td><td>The (A) division starts as the zygote moves through the isthmus of the oviduct called (B).</td><td>(A) Mitotic
(B) Cleavage</td></tr>
<tr><td>80.</td><td>The embryo with 8 to 16 blastomeres is called ?</td><td>Morula</td></tr>
<tr><td>81.</td><td>The blastomeres in the blastocyst are arranged into an outer layer called (A) and an inner group of cells attached to trophoblast called (B).</td><td>(A) Trophoblast
(B) Inner cell mass</td></tr>
<tr><td>82.</td><td>Some finger like projections appear on the trophoblast called ?</td><td>Chorionic villi</td></tr>
<tr><td>83.</td><td>The structural and functional unit between developing embryo and maternal body called ?</td><td>Placenta</td></tr>
<tr><td>84.</td><td>The placenta is connected to the embryo through an (A).</td><td>Umbilical cord</td></tr>
<tr><td>85.</td><td>Placenta acts as an .....</td><td>Endocrine tissue</td></tr>
<tr><td>86.</td><td>Which of the following hormones are not secrete by placenta ?
HCG, hPL, progesterone, oxytocin, estrogen, prolactin, relaxin</td><td>Oxytocin, prolactin</td></tr>
<tr><td>87.</td><td>Match the column :-
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>(a)</td><td>After one month of pregnancy</td><td>(i)</td><td>Heart is formed</td></tr>
<tr><td>(b)</td><td>End of second month of pregnancy</td><td>(ii)</td><td>Major organ systems are formed</td></tr>
<tr><td>(c)</td><td>End of first trimester of pregnancy</td><td>(iii)</td><td>First movement of foetus</td></tr>
<tr><td>(d)</td><td>During the fifth month of pregnancy</td><td>(iv)</td><td>Limbs and digits develop</td></tr>
</table></td><td>a-(i), b-(iv), c-(ii), d-(iii)</td></tr>
<tr><td>88.</td><td>Parturition is induced by a complex neuroendocrine mechanism. (True/false)</td><td>True</td></tr>
<tr><td>89.</td><td>The signals for parturition originate from the fully developed foetus only. (True/false)</td><td>False</td></tr>
<tr><td>90.</td><td>Which hormone acts on the uterine muscle and causes stronger uterine contractions.</td><td>Oxytocin</td></tr>
<tr><td>91.</td><td>The milk produced during the initial few days of lactation is called ?</td><td>Colostrum</td></tr>
</table>

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Reproductive Health

<table>
<tr><th>Q.NO.</th><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>1.</td><td>Which country in world to initiate family planning programme in 1951.</td><td>INDIA</td></tr>
<tr><td>2.</td><td>Improved programmes covering wider reproduction-related areas are currently in operation under the popular name.</td><td>'Reproductive and child Health care (RCH) programmes.</td></tr>
<tr><td>3.</td><td>Amniocentesis is statutory banned for</td><td>Sex determination</td></tr>
<tr><td>4.</td><td>Central Drug Research Institute (CDRI). Scientists developed a new oral contraceptive for the females is.</td><td>Saheli</td></tr>
<tr><td>5.</td><td>Periodic abstinence is which type of contraceptive method</td><td>Natural method</td></tr>
<tr><td>6.</td><td>In periodic abstinence couple avoid coitus from</td><td>10 to 17 day of menstrual cycle</td></tr>
<tr><td>7.</td><td>In menstrual cycle day 10 to 17 is known as</td><td>Fertile period</td></tr>
<tr><td>8.</td><td>Lactational amenorrhea is effective only upto maximum period of</td><td>6 month</td></tr>
<tr><td>9.</td><td>'Nirodh' is a popular brand of</td><td>Condom for male</td></tr>
<tr><td>10.</td><td>Which contraceptive method also prevent from STD and AIDS ?</td><td>Condom</td></tr>
<tr><td>11.</td><td>Match the columns.
<table>
<tr><th></th><th>Column-A</th><th></th><th>Column-B</th></tr>
<tr><td>a</td><td>Non-medicated IUDs</td><td>i</td><td>Uterus unsuitable for implantation</td></tr>
<tr><td>b</td><td>Copper-releasing IUDs</td><td>ii</td><td>Inhibit ovulation</td></tr>
<tr><td>c</td><td>Hormone-releasing IUDs</td><td>iii</td><td>Cutting of fallopian tube</td></tr>
<tr><td>d</td><td>Progesterone-pill</td><td>iv</td><td>Suppress motility of sperms</td></tr>
<tr><td>e</td><td>Tubectomy</td><td>v</td><td>Increases phagocytosis of sperms</td></tr>
<tr><td>f</td><td>Vasectomy</td><td>vi</td><td>Cutting of vas-deferens</td></tr>
</table></td><td>a-v, b-iv, c-i, d-ii, e-iii, f-vi</td></tr>
<tr><td>12.</td><td>Ideal contraceptives for the females who wants to delay pregnancy.</td><td>IUDs</td></tr>
<tr><td>13.</td><td>Oral pills for females causes.</td><td>Inhibit ovulation and implantation</td></tr>
<tr><td>14.</td><td>Once a-week oral pill is :</td><td>Saheli</td></tr>
<tr><td>15.</td><td>In emergency which contraceptive methods may be used within 72 hours of coitus.</td><td>IUD & progestogen or progestogen estrogen combination pills.</td></tr>
<tr><td>16.</td><td>Which contraceptive method are also known as sterilisation ?</td><td>Surgical method</td></tr>
<tr><td>17.</td><td>Sterilisation in male is known as :</td><td>Vasectomy</td></tr>
<tr><td>18.</td><td>Government of INDIA legalised MTP in.</td><td>1971</td></tr>
<tr><td>19.</td><td>MTP is considered relatively safe during.</td><td>First trimester</td></tr>
</table>

20.	In ZIFT-early embryo is transferred in (upto 8 blastomeres)	Fallopian tube
21.	Full form of ZIFT	Zygote Intra Fallopian Transfer
22.	In IUT, embryos with more than 8 blastomeres are transferred in	Uterus
23.	Full form of GIFT	Gamete Intra Fallopian Transfer
24.	Procedure to form an embryo in the laboratory in which a sperm is directly injected into the cytoplasm of ovum	ICSI-Intracytoplasmic sperm injection
25.	If male is unable to inseminate the semen into vegina then which procedure is used.	Artificial insemination (AI) or IUI-Intrauterine insemination.

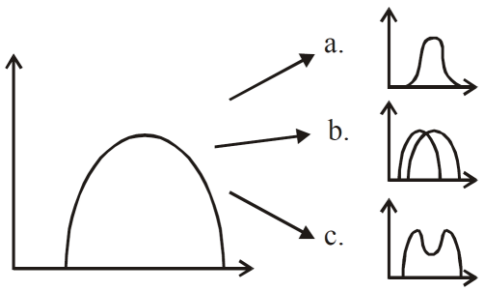
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Evolution

Q.NO.	QUESTION	ANSWER
1.	The study of history of life forms on earth is known as :	Evolutionary Biology
2.	Stellar distances are measured in.	Light years
3.	The universe is very old almost.	20 Billion years
4.	Which theory attempts to explain us the origin of universe?	Big-Bang Theory
5.	In the solar system of the milky way galaxy, the earth was supposed to have been formed about :	4.5 Billion years back
6.	Life come out of decaying and rotting matter like straw, mud etc. This was the theory of :	Spontaneous generation
7.	The scientist who by careful experimentation demonstrated that life comes only from pre-existing life :	Louis Pasteur
8.	Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules (e.g. RNA, protein etc.) and that formation of life was preceded by :	Chemical evolution
9.	In 1953, S.L. Miller, an American scientist, created electric discharge in a closed flask containing which gases at 800°C.	CH_4, H_2, NH_3 and water vapour
10.	Based on observations made during a sea voyage in a sail ship called H.M.S. Beagle around the world who concluded that existing living forms share similarities :	Charles-Darwin
11.	The fitness, according to Darwin, refers ultimately and only to	Reproductive fitness
12.	According to Darwin, mechanism of evolution is	Natural selection
13.	Remains of hard parts of life forms found in rocks are	Fossils
14.	Study of fossils is known as :	Paleontology
15.	The same structure developed along different directions due to adaptions to different needs. These structures are called as :	Homologous structure
16.	Analogous structures are a result of :	Convergent evolution
17.	The flippers of Penguins and Dolphins are an example of :	Analogy
18.	Which can be used as industrial pollution indicators?	Lichens
19.	Moths that were able to _____ themselves. i,e hide in the background, survived.	Camouflage
20.	We employ antibiotics or drugs against eukaryotic organisms/cells. Hence, resistant organisms/cells are appearing in a time scale of month or years and not centuries. These are example of evolution due to	Anthropogenic -action
21.	On which island Darwin observed many varieties of finches?	Galapagos Islands
22.	Who worked in Malay Archipelago?	Alfred Wallace

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23.	The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called :	Adaptive radiation
24.	Two examples of adaptive radiations are :	Darwin's finches and Australian- marsupials
25.	When more than one adaptive radiations appeared to have occurred in an isolated geographical area (representing different habitats). One can call this :	Convergent evolution
26.	The rate of appearance of new forms is linked to the life cycle or the life span. (True/False)	True
27.	The geological history of earth closely correlates with the biological history of earth. (True/False)	True
28.	According to Darwin, adaptive ability is acquired. True/False	False (It is inherited)
29.	Two key concepts of Darwinian theory of evolution are :	Branching descent and natural selection
30.	The work of Thomas Malthus on populations influenced :	Darwin
31.	Hugo de Vries based on his work on which plant brought forth the idea of mutation ?	Evening-primrose
32.	Single step large mutation is called :	Saltation
33.	Which principle says that allele frequencies in a population are stable and are constant generation to generation. $p^2 + 2pq + q^2 = 1$	Hardy Weinberg principle
34.	In the given diagrammatic representation of operation of natural selection, identify a, b and c 	(a) Stabilising (b) Directional (c) Disruptive
35.	Natural selection can lead to stabilisation, directional change or disruption. True/False	True
36.	The first cellular forms of life appeared on earth about _____ million years ago (mya)	2000
37.	Jawless fish probably evolved around 500 mya. (True/False)	False (350 mya)
38.	In 1938, a fish caugnt in _____ happened to be a coelacanth which was thought to be extinct.	South Africa
39.	_____ evolved into the first amphibians.	Lobefins

40.	_____ were present but they all fell to form coal deposits slowly.	Giant ferns (Pteridophytes)
41.	About 200 mya the dinosaurs suddenly disappeared from the earth. (True/False)	False (It is 65 mya)
42.	*Dryopithecus* and *Ramapithecus* were existing about_____.	15 mya
43.	Two mya, which man ancestor probably lived in East African grasslands?	Australopithecines
44.	*Homo habilis* has the brain capacities about.	650-800 c.c
45.	*Homo erectus* had a large brain around 900 c.c and probably ate meat. (True/False)	True
46.	Who with a brain size of 1400 cc lived in near east and central Asia between 1,00000–40,000 years back.	Neanderthal man
47.	*Homo sapiens* arose in which place and moved across continents and developed into distinct races.	Africa
48.	The story of evolution of modern man is most interesting & appear to parallel evolution of human brain and language. (True/ False)	True
49.	Match the column. a. Ice age b. Pre historic cave art c. *Australopithecus* d. Fish like reptile e. Biggest dinosaurs f. *Homo erectus* g. Therapsids h. Coelacanth i. Darwinian variations j. Mutations i. East Aftrican grasslands ii. *Ichthyosaurs* iii. *Tyrannosaurus* iv. 18000 yrs ago v. 75000–10000 yrs ago vi. Mammal ancestor vii. 1.5 mya viii- Small and directional ix. random and directionless x. Lobefins	a. v b. iv c. i d. ii e. iii f. vii g. vi h. x i. viii j. ix
50.	The correct sequence of human ancestors in increasing brain capacitiy. a. Neanderthal man b. *H. erectus* c. *H. habilis*	c, b, a
51.	Big bang, a single huge explosion causes __A__ of universe and hence temperature __B__.	A = Expansion B = Came down
52.	First formed gases in course of origin of universe.	Hydrogen and Helium
53.	Molten masses covered the surface of earth act as source of A, B, C and D during the formation of atmosphere.	A - Water; B - Methane; C - Carbondioxide; D - Ammonia

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54.	Early Earth has no	Atmosphere
55.	Ozone formed in upper atmosphere i.e. stratosphere by the action of_____ on O_2.	UV radiations
56.	Life originated in _____ and cannot sustain without_____.	Water, Water
57.	Life appeared _____ after the formation\origin of earth i.e. almost 4 bya.	500 mya
58.	Limitation of Louis pasteur theory was that it cannot explain the origin of ___	Ist form of life
59.	Louis pasteur used _____ as a culture medium in his famous experiment (Swan nack flask experiment) and disproved the Theory of spontaneous generation i.e. life originated spontaneously from dead and decay matter, mud etc.	Killed yeast+sugar solution
60.	Uninterrupted source of energy (SUN), reducing environment (CH_4, etc. and absence of O_2), water and self replicating organic material required for _____.	Origin of life
61.	In an experiment in 1953 SL Miller proved chemical evolution by forming _A_ i.e. organic materials from inorganic materials (NH_3, H_2O, H_2 CH_4) at high temperature in reducing environment while other scientists observed B, C, D and E i.e. organic materials in similar experiments like Miller.	A = amino acids ; B = Sugars ; C = Nitrogen Bases ; D = Pigments ; E = Fats
62.	_____act as an evidence to establise similarity in the process of chemical evolution on earth and elsewhere in space	Meteorite
63.	3 bya gaint molecules like RNA proteins polysaccharides etc. form first _____ which probably show reproduction.	Non cellular forms of life
64.	First cellular form of life was_____	unicellular
65.	Natural selection i.e.__(A)____ acts on__(B)__ and selection is based on __(C)__not genotype	mechanism of evolution, population, phenotype
66.	_____is the end result of the ability to adapt and get selected by nature.	Fitness
67.	Evolution of life forms driven by use and disuse of organs.	Lamarck
68.	Evolution is a process when we talk about the__(A)__and result of a process when we talked about the story of life on earth.	A = Story of this world
69.	Evolution by natural selection in true sense started when cellular forms of life with _____ in metabolic capability originated on earth.	Differences
70.	Fitness is based on characteristics which are _____ i.e. a genetic basis for getting selected and to evolve.	inherited
71.	Different forms of placental mammals in Australia are an example of	Adaptive radiation or divergent evolution

72.	Darwin's finches i.e. small black coloured birds are originally____ but due to divergent radiation adapted to various food sources like insects, fruits etc.	seed eaters
73.	Fossils provide important evidence for _____ and found or formed in sedimentary rocks.	Evolution
74.	Certain features during embryonic state common to all vertebrates are absent in adult.	Ernst Hackel
75.	Embroys never pass through the adult stages of other animals and hence disproved Heckel's theory.	Karl Ernst Von Baer
76.	Homology indicates	Common ancestory
77.	Anthropogenic actions leads to evolution by natural selection are __A__ and __B__ while artificial selection by __C__.	A = Industrialisation B = Application of antibiotics, Herbicides and pesticides C = Animal and plant breeding programs
78.	Change in allele frequencies or Hardy Weinberg equilibrium or genetic equilibrium of a randomly mating large population results in :-	Evolution
79.	Genetic equilibrium of a randomly mating large population can be altered by.	Gene migration/Gene flow, Genetic drift, Mutation, genetic recombination and natural selection
80.	Skull of adult _____ is more similar to skull of baby chimpanzee than adult.	Modern man / Adult human
81.	In nature evolution is not a directed process in the sense of_____	determinism
82.	Evolution is a _____based on chance events in nature and chance mutations in the organism.	stochastic process
83.	Whales, Dolphins, seals and sea cows	aquatic mammals.
84.	Evolution of language and _____ enhances the success of humans.	self consciousness
85.	Consideration of only those variations which were heritable and makes best utilization of resources or makes individual best adaptive to habitat i.e. responsible for more progeny_____.	Novelty and brilliant insight of Darwin
86.	Mutations (directionless or random) cause large differences in population i.e. cause speciation or evolutions in single step (saltation).	Hugo de vries
87.	Exact mechanism of how non-cellular aggregation could evolve into cells with membranous envelop is	not known

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88.	Dryopithecus was more __(A)__ while ramapithecus was more __(B)__.	A = ape like, B =man like
89.	First human like being, the hominid probably not eat meat.	Homo habilis.
90.	Human ancestor used hides to protect their body and burried their dead.	Neanderthalman
91.	Bones found in Ethiopia and Tanzania revealed that about __A__ man - like primate not taller than __B__ but __C__ right.	A = 3-4 mya ; B = 4 feet ; C = Walked up
92.	First organism that invade land	Plants
93.	Reptiles lay ____ which do not dry up in sun unlike those of amphibians.	thick shelled eggs
94.	Origin of invertebrates __(A)__, sea weeds and few plants__(B)__and fish with stout and strong fins could move on land and go back to water __(C)__.	A = 500 mya B = 320 mya C = 350 mya
95.	First mammals were like _____.	shrews
96.	_____ dominated on earth after dinosaurs.	Mammals
97.	Pouched mammals of Australia survived because of lack of competition from any other mammals due to _____.	Continental drift
98.	Similarities in proteins and genes performing a given function among diverse organisms gives clues to ____ like structural similarities.	common ancestory

Human Health and Disease

Q.NO.	QUESTION	ANSWER
1.	In early times Hippocrates & Indian Ayurveda system asserted that state of body and mind are balanced by :-	Humors
2.	"Good Humor" hypothesis was given by :-	Hippocrates & Ayurveda
3.	"Good Humor" hypothesis was discarded by :-	William Harvey
4.	Health is affected by :-	Genetic disorder, infections & life style
5.	Health is maintained by :-	Balance diet, personal hygiene & regular exercise
6.	Health brings :-	Productivity, economic prosperity, longevity of people & decreased mortality of mother & infant.
7.	Adversely affected organs characterised by signs and symptoms:-	Disease
8.	Disease which are transmitted from one person to another is known as :-	Infectious disease
9.	Major cause of death among non-infectious disease is :-	Cancer
10.	Disease causing agents are known as :-	Pathogens
11.	Due to interference of pathogen in normal vital activities results:-	Morphological & functional damage of cell
12.	For survival of pathogen within the environment of the host, pathogen have to :-	Adapt to life within the environment of the host.
13.	Transmission of "*Salmonella typhi*" through :-	Contaminated food & water
14.	Symptoms like sustained high fever (39° to 40°C), weakness, stomach pain, constipation, headache & loss of appetite are related with :-	Typhoid
15.	Death may occur in severe case of typhoid due to :-	Intestinal perforation
16.	Confirmatory test for typhoid is :-	Widal
17.	Carrier of typhoid, who spread disease for several years was:-	Mary Mallon
18.	Bacteria responsible for pneumonia is :-	*Streptococcus pneumoniae* / *Haemophilus influenzae*
19.	Site of infection in pneumonia :-	Alveoli of lungs
20.	Symptoms like fever, chills, cough, headache and in severe case turning of color of nail from gray to bluish occurs in :-	Pneumonia
21.	Malaria is a _____ disease.	Protozoan
22.	The most serious type of malaria, malignant malaria which can even be fatal is caused by :-	*Plasmodium falciparum*
23.	Infectious form of *Plasmodium* for humans is :-	Sporozoites

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24.	Transmitting agent for *Plasmodium* species is :-	Female *Anopheles* mosquito
25.	Toxin in malaria, which results in chill & high fever in humans is:-	Haemozoin
26.	The storage area for *Plasmodium* in female *Anopheles* mosquito is :-	Salivary gland
27.	Sexual stage which develops in RBCs is :-	Gametocytes
28.	Causative agent of a protozoan disease which cause amoebiasis is :-	*Entamoeba histolytica*
29.	Mechanical carrier for amoebic dysentery is :-	House fly
30.	Common round worm, which shows symptoms like internal bleeding, muscular pain, fever, anemia & blockage of intestinal passage is :-	*Ascaris*
31.	Filarial worm which cause chronic inflammation in lymphatic vessels, usually in lower limbs is :-	Elephantiasis or filariasis
32.	Genera which are responsible for fungal disease ringworm are:-	*Microsporum, Trichophyton* & *Epidermophyton*
33.	Dreadful disease which is completely eradicated is :-	Small pox
34.	Non specific type of defense that is present at the time of birth is :-	Innate immunity
35.	The main physical barrier of our body which prevents entry of micro-organism from environment is :-	Skin & mucosal epithelium
36.	Polymorpho-nuclear leucocytes (PMNL) which form cellular barrier are :-	Neutrophils
37.	Protein which is secreted by virus infected cell and form cytokine barrier is :-	Interferon
38.	When our body encounters first time with a pathogen. Then the first time low intensity response is known as :-	Primary response
39.	Highly intensified second time response in acquired immunity on the basis of previous memory is called as :-	Anamnestic response (secondary)
40.	Proteins which are produced by B-lymphocytes in response to pathogen are :-	Antibodies
41.	Formula which represent the antibody is :-	H_2L_2
42.	Antibody mediated immunity & cell mediated immunity are represented respectively by :-	B-lymphocyte & T-lymphocyte
43.	When the immune system of host are exposed to antigen and produces antibodies it is called as :-	Active immunity
44.	Yellowish fluid which contains IgA antibody and protects infants in initial days :-	Colostrum
45.	Principle of vaccination or immunisation is based on :-	Memory
46.	When preformed antibodies against the antigen are injected to patient and immunity is acquired than type of immunity is:-	Passive immunity

47.	Vaccine produced using method of recombinant DNA technology is :-	Hepatitis-B
48.	Exaggerated response of immune system to certain antigens is known as :-	Allergy
49.	Due to genetic or unknown reasons, the immune system attacks on self cell, it is called as :-	Auto immunity
50.	Constitutes 50% of lymphoid tissue and located within the major respiratory, digestive & urogenital tract is known as :-	Mucosa associated lymphoid tissue (MALT)
51.	Enzyme which is required in AIDS disease to form viral DNA from viral RNA is :-	Reverse transcriptase
52.	Number of these cells decreases progressively during HIV infection leading to immunodeficiency.	Helper T-lymphocytes
53.	Infections occurring in AIDS due to immunodeficiency are caused by:-	*Mycobacterium*, viruses, fungi and parasites like *Toxoplasma*.
54.	Widely used diagnostic test for AIDS is :-	ELISA
55.	The cell which continues to produce virus and act like HIV factory is :-	Macrophage
56.	Phenomenon of cell by which it inhibits the uncontrolled growth is :	Contact inhibition
57.	The property of malignant tumor through which it spreads all over body is :-	Metastasis
58.	Physical, chemical or biological factors which transform a normal cell into cancerous cell, known as :-	Carcinogen
59.	Biological response modifier which activate immune system and help in destroying the tumor is :-	α-interferon
60.	White, odorless, bitter crystalline compound which are chemically diacetyl morphine is :-	Smack/Heroin
61.	Receptor for the opioid drugs are present in our body at :-	Central Nervous system and gastrointestinal tract
62.	Marijuana, hashish, charas and ganja are examples of :-	Cannabinoids
63.	Native to South America which interfere with the transport of dopamine neurotransmitter is :-	Coca alkaloid or cocaine
64.	The first use of drug may be out of :-	Curiosity or experiment
65.	Anxiety, shakiness, nausea and sweating are relieved when use is resumed are known as :-	Withdrawal syndrome
66.	Reckless behavior, vandalism violence etc. are immediate adverse effect of :-	Drug & alcohol abuse

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Biotechnology and Its Application

Q.NO.	QUESTION	ANSWER
1.	Which technology is used to purify the protein / organic	Downstream processing
2.	Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called _____.	Genetically modified organism (GMO)
3.	Which of the following statement is true / false. (i) Genetic modification has made crops more tolerant to abiotic stress. (ii) Genetic modification has increased reliance on chemical pesticides. (iii) Agrochemical based agriculture is a option that can be thought for increasing food production. (iv) The green revolution succeeded in tripling the food supply and it was enough to feed growing human population.	 True False True False
4.	Which type of plants are created with the use of GM to supply alternative resources to industries?	Tailor made plants
5.	Bt toxin is produced by a bacterium called _____.	*Bacillus thuringiensis*
6.	Which type of gene has been cloned from the bacteria and has been expressed in the plant to provide resistance to insects.	Bt toxin gene
7.	Name the orders of insects which are killed by proteins that is produced by some strains of *Bacillus thuringiensis* :-	Lepidopterans, Coleopterans and Dipterans
8.	Which type of pH is responsible for conversion of inactive Bt protoxin into active form of toxin	Alkaline pH
9.	Fill in the blanks : (i) The activated Bt toxin binds to the ___(a)___ and create ___(b)___. (ii) *B.thuringiensis* form a protein crystal during a particular ___(a)___ these crystals contain a toxic ___(b)___. (iii) The choice of Bt genes depends upon ___(a)___ and ___(b)___ as most Bt toxins are ___(c)___.	(a) Midgut epithelial cell (b) Pores (a) Phase of their growth (b) Insecticidal protein (a) The crop (b) the target pest (c) Insect group specific
10.	What is the name of that gene which is responsible for the coding of Bt toxin	*cry* gene
11.	Give the two name of cry gene which control the cotton bollworms :-	*cry I Ac, cry II Ab*
12.	Which nematode infects the roots of tobacco plants	Meloidegyne incognitia
13.	A novel strategy was adopted to prevent the infection of nematode was based on which process?	RNA interference (RNAi)

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14.	Which mechanism takes place in all eukaryotic organisms as a method of cellular defense	RNAi
15.	In RNA the source of complementary RNA could be from an infection by viruses having RNA genomes or Mobile genetic elements (transposons) that form complementary RNA replicate via an _____.	RNA intermediate
16.	Match the column : **Column-A** **Column-B** (1) RNAi (a) Corn borer (2) *cry I Ab* (b) Cotton bollworms (3) *cry I Ac* (c) Proinsulin (4) C-peptide (d) Silencing of specific mRNA	1-d, 2-a, 3-b, 4-c
17.	In earlier insulin used for diabetes was extracted from pancreas of which animals?	Slaughtered cattle and pigs
18.	In mature insulin chain-A and chain-B are linked together by which bond?	Disulphide bridges
19.	Which peptide chain is removed during maturation of proinsulin	C-Peptide
20.	What is the main challenge for production of insulin using r-DNA techniques	Getting insulin assembled into a mature form.
21.	If a person is born with a hereditary disease, which corrective therapy be taken for such a disease	Gene therapy
22.	The first clinical gene therapy was given in _____.	1990
23.	Adenosine deaminase enzyme is crucial for which system to function?	Immune system
24.	Give the name of any three techniques that serve the purpose of early diagnosis	r-DNA technology, ELISA & PCR
25.	Very low concentration of a bacteria or virus (at a time when the symptoms of the diseae are not yet visible) can be detected by	Amplification of their nucleic acid by PCR
26.	Which technique is now routinely used to detect HIV in suspected AIDS patients?	PCR
27.	ELISA is a molecular diagnosis technique is based on the which principle?	Antigen antibody interaction
28.	Animal that have had their DNA manipulated to possess and express an extra (foreign) gene are known as :-	Transgenic animals
29.	Human protein α-1-antitrypsin used to treat _____.	Emphysema
30.	Name of the first transgenic cow is :-	Rosie

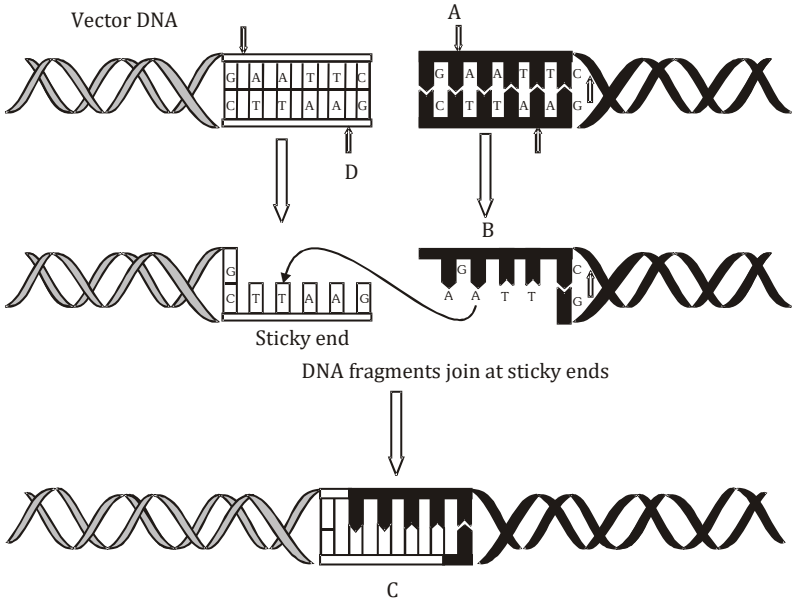
31.	The milk of first transgenic cow contained a human protein ____.	Human alpha-lactalbumin
32.	Which organisation in India make decisions regarding the validity of GM research and safety of introducing GM organisms for public services?	GEAC (Genetic engineering approval committee)
33.	The use of bioresources of multinational companies and other organisation without proper authorisation from the countries and people concerned without compensatory payment is called :-	Biopiracy
34.	The first transgenic cow, Rosie, produced human protein enriched milk which contained_____, by weight its is _____.	alpha lactalbumin ; 2.4 gm/*l*
35.	The proteins coded by the genes _____ and _____ control the cotton bollworms and corn borer, respectively.	Cry II Ab, Cry I Ab
36.	About _____ recombinant therapeutics have been approved for human use the world over. In India_____ of these are presently marketed.	30, 12
37.	Gene therapy is used to treat diseases especially _____diseases.	Hereditary disease
38.	The activated Bt toxin binds to the surface of _____cells.	Midgut, Epithelial
39.	Gene therapy is a collection of method that allows correction of ____defect that has been diagnosed in an _____.	Gene, Embryo
40.	Nutrient media for plant tissue culture consists of the following :	Carbon source, growth regulators, vitamins etc.
41.	RNAi prevents :	Translation of mRNA
42.	Meristem is grown in vitro to produce virus free plants because.	They are activity dividing cells
43.	Pomato is formed by the process known as _____.	Somatic hybridization

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Biotechnology : Principles and Processes

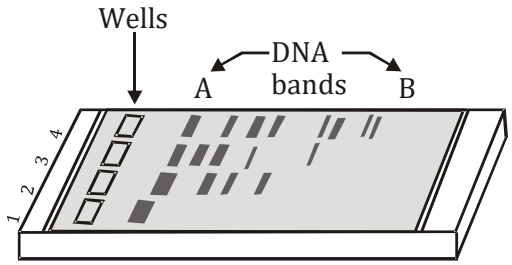
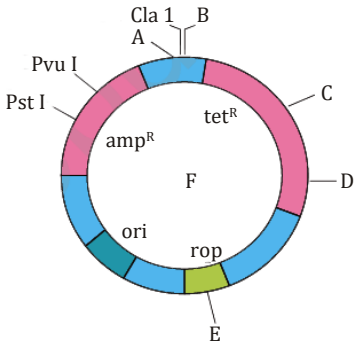
Q.NO.	QUESTION	ANSWER
1.	The techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.	Biotechnology
2.	Who defined biotechnology as the integration of natural science and organisms cells, parts thereof, and molecular analogues for products and services?	European Federation of Biotechnology (EFB)
3.	Write down the two core techniques that enabled birth of modern biotechnology.	(i) Genetic engineering (ii) Bioprocess engineering
4.	Asexual reproduction preserves the _____, while sexual reproduction permits _____.	Genetic information, variation
5.	Which techniques of genetic engineering overcome the limitation and allows us to isolate and introduce only one or a set of desirable genes without introducing undersirable genes into the target organism ?	Creation of recombinant DNA, gene cloning and gene transfer
6.	The fate of piece of DNA, which is somehow transferred into an alien organism would always integrate and multiply. (True/False)	False
7.	Name the specific sequence which is responsible for initiation of replication?	Origin of replication (ori.)
8.	The technique which is responsible for making multiple identical copies of any template DNA called _____.	Cloning
9.	Autonomously replicating circular extra-chromosomal DNA in bacterial cell is called _____.	Plasmid
10.	Who constructed the first recombinant DNA ?	Stanley Cohen and Herbert Boyer (1972)
11.	Which enzyme is known as 'molecular scissors' ?	Restriction endonuclease
12.	The cutting of DNA at specific locations is become possible due to _____.	Restriction endonuclease (Restriction enzyme)
13.	When a cut piece of DNA linked with plasmid and transfers the linked DNA into host cell then in this condition plasmid act as ___.	Vector
14.	Mosquito acts as an _____ to transfer _____ the into human body.	Vector, malarial parasite
15.	Which enzyme links the antibiotic resistance gene with the plasmid vector ?	DNA ligase
16.	When a cut piece of DNA linked with plasmid with the help of ligase enzyme then this result in formation of new DNA, known as _____.	Recombinant DNA (r-DNA)

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17.	What are the three basic steps in genetically modifying an organism?	1. Identification of DNA with desirable genes 2. Introduction of the identified DNA into the host 3. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.
18.	Name the enzyme that cut the DNA at specific site :-	Restriction endonuclease
19.	The first discovered restriction endonuclease enzyme was :-	*Hind II*
20.	Hind-II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs. This specific base sequence is known as the _____.	Recognition sequence
21.	Today we know more than _____ that have been isolated from over _____ of bacteria each of which recognize difference recognition sequences.	900 restriction enzymes, 230 strains
22.	Match the column for enzyme EcoRI. **Column A** **Column B** 1. E A. Order of enzyme isolation 2. co B. genus 3. R C. name of strain 4. I D. species	1–B, 2–D, 3–C, 4–A
23.	Restriction enzymes belong to which class of enzymes ?	Nucleases
24.	Which nuclease remove nucleotides from the ends of the DNA ?	Exonuclease
25.	Which specific nucleotide sequences is recognizes by restriction endonuclease in DNA?	Palindromic nucleotide sequences.
26.	What is the Recognition sequence of EcoRI ?	5'GAATTC3'
27.	Identify the A, B, C and D in given diagram.  	A- Foreign DNA B- Sticky end C- Recombinant DNA D- EcoRI

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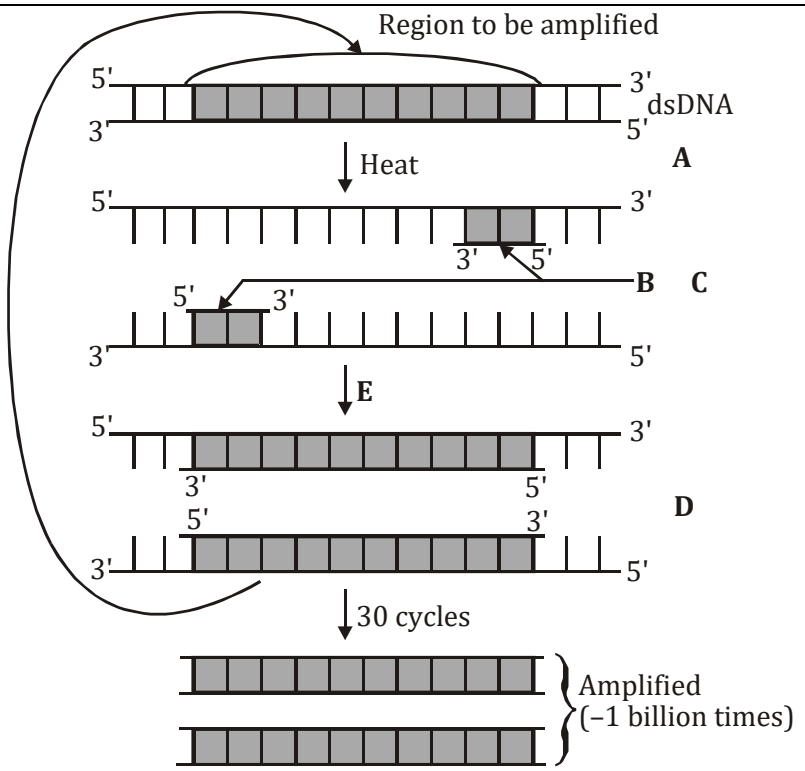
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28.	The groups of letters that form the same words when read both forward and backward, e.g. "MALAYALAM" is called _____.	Palindromes
29.	What is the process of insertion of recombinant DNA in bacterial cell?	Transformation
30.	The cell which contain recombinant DNA.	Transformed and recombinant cell
31.	Name the technique which is use to separate the DNA fragments?	Gel electrophoresis
32.	In gel electrophoresis which compound is use to stain DNA fragments.	Ethidium bromide
33.	Identify the A and B in given diagram.  	A- Largest, B-Smallest DNA fragment
34.	Now a days which is the most commonly used matrix for gel electrophoresis ?	Agarose
35.	From where the agarose is extracted ?	Sea weeds
36.	Name the useful selectable markers for *E.coli* ?	Ampicillin, chloramphenicol, tetracycline or kanamycin, resistance gene
37.	Identify the A, B, C, D, E and F in the given diagram.  	A-EcoRI, B-Hind III, C-BamH I, D-Sal I, E-Pvu II and F-pBR322
38.	The extraction of DNA fragment from agarose gel is called _____.	Elution
39.	Match the column. **Column A** / **Column B** 1. Recombinant DNA technology — A. Vector 2. Cloning vehicle — B. Joining enzyme 3. Macromolecule separation — C. Gel electrophoresis 4. DNA ligase — D. Genetic engineering	1-D, 2-A, 3-C, 4-B

40.	Name the process which is use to check the progression of restriction enzyme digestion.	Agarose gel electrophoresis
41.	Which type of substrate produce colour for differentiation of recombinant and non-recombinant colonies ?	Chromogenic substrate
42.	True and false : 1. The uptake of DNA during transformation is an active process. 2. In recombinant DNA, DNA is inserted within the coding sequence of an enzyme β galactosidase. 3. Bioreactor provides the optimal conditions for obtaining desired product. 4. In insertional inactivation process, alien gene is introduced in ori. sequence.	 True True True False
43.	The process include separation and purification of desirable protein is called _____.	Downstream processing
44.	After insertion inactivation of *β - galactosidase gene* which type of colonies do not produce any colour ?	Recombinant colonies
45.	The part of Ti Plasmid is transferred by *Agrobacterium* in dicot plant is called.	T DNA
46.	If any protein encoding gene is expressed in a heterologus host, is called a _____.	Recombinant protein
47.	Which type of ion use to increase the efficiency of uptake of DNA by bacterial cell ?	Divalent cation (Ca^{++})
48.	The bacterial cell which is able to uptake plasmid DNA is called _____.	Competent cell
49.	In biolistic or gene gun method, DNA is coated on _____.	Gold or Tungsten microparticles
50.	By which method rDNA is directly injected into the nucleus of animal cell ?	Microinjection
51.	Which enzyme is use for removal of RNA molecule in DNA isolation process.	Ribonuclease
52.	In DNA isolation for the precipitation reaction which chemical is used?	Chilled ethanol
53.	Identify A, B, C, D, E in given diagram	A-Denaturation, B-Primers, C-Annealing, D-extension, E-Taq Polymerase and deoxynucleotides

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54.	From which bacteria thermostable taq DNA polymerase is isolated ?	*Thermus aquaticus*
55.	The first recombinant DNA was made using native plasmid of _____ that could be transformed into _____.	*Salmonella typhimurium, E. coli*
56.	The first discovered restriction endonuclease is :-	Hind II
57.	Bright orange coloured bands of ____ are seen in an ethidium bromide stained gel exposed to UV light.	DNA
58.	The definition of biotechnology "The integration of _____ and organisms, cells, parts thereof and _____ for products and services".	Natural science, molecular analogues
59.	If any protein encoding gene is expressed in a heterologous host, it is called a _____ protein.	Recombinant
60.	Presence of insert results into insertional inactivation of the β-galactosidase gene and the colonies are _____ and they are identified as _____.	Colourless, Recombinant
61.	EcoRI comes from :-	*Escherichia coli RY13*
62.	Bright orange colored bands of DNA are seen in a _____ stained gel exposed to UV light.	Ethidium bromide
63.	A _____ bioreactor is used in biosynthesis stage.	stirred tank
64.	Arrange the following steps of recombinant DNA technology. (a) Extraction of desired product. (b) Transferring the recombinant DNA into host. (c) Fragmentation of DNA by restriction endonuclease. (d) Isolation of a desired DNA fragment (e) Culturing the host cells in a medium at large scale (f) Ligation of the DNA fragment into a vector (g) Isolation of DNA	g, c, d, f, b, e, a